



Live Models: Integrating Prediction Markets into AMTAIR's Bayesian Risk Networks

“Managing Contamination Loops in Collaborative Forecasting Systems”

Abstract:

AMTAIR (Automated Modeling of Transformative AI Risks; V. J. Meyer (2025a)'s thesis project) extracts Bayesian networks from natural language AI safety sources to formalize risk arguments. These extracted models embed the epistemic state of authors at writing time, a significant limitation in a domain where capabilities evolve monthly. Prediction markets offer continuously updated probability estimates from distributed intelligence, and Berea et al. (2012) have demonstrated that Bayesian network structure can organize such forecasts. But naïve integration risks contamination loops: if AMTAIR publishes estimates that forecasters observe, re-ingesting their subsequent forecasts as “independent evidence” produces spurious confirmation: uncertainty laundered through markets returning as false validation.

This paper argues that the contamination risk is constitutive of AMTAIR's design: core properties the system needs to fulfill its purpose (publicity, authority, broad uptake) are also properties that create contamination risk. I outline a methodology combining hierarchical Bayesian inference, randomized temporal embargoes, and quality-weighted multi-platform aggregation, but argue that these statistical tools require organizational infrastructure to function. They rely on inputs they do not generate (organizational facts about information flows), operate through channels they do not model by default (authority, deference, structural impediment), and depend on conditions they do not create (epistemic culture, institutional commitment). Organizational epistemology, specifically the analysis of epistemic vices and virtues at the collective level, provides an analytical framework for specifying implied infrastructure requirements. The paper concludes that managed integration, with named vulnerabilities and organizational safeguards designed to maintain the conditions statistical tools require, is categorically preferable to the unmanaged integration that will occur regardless.

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I. Introduction: The Integration Trap

I.1 Opening Vignette: AI Governance Decisions

A biosecurity red-team report of a model scheduled for open-weights release, leaked forty-eight hours ago, shows a frontier AI system guiding the synthesis of a novel pathogen variant, the first credible evidence that a soon to be open-sourced language model unlocks bio-terrorism for nefarious groups. A governance analyst who spent the past quarter producing AMTAIR-derived risk assessments for her agency's AI oversight program¹ has to decide whether to recommend emergency restrictions on releasing the weights of the model. The correct response depends on what actually happened.²

AMTAIR decomposes the question across five main causal hypotheses (de novo synthesis capability, recombinative accessibility, evaluation elicitation artifact, training data memorization, scaffolding-contingent capability) with posterior probabilities and confidence intervals for each.³ On the central question for her brief, *Does this constitute genuine bio-uplift?*, the ensemble reads 35% [20%, 52%].

But Metaculus shows 72%. Manifold Markets at 68%. AMTAIR's decomposition comes with *explainability*: the analyst can trace for policymakers *how* and *why* 35%, which sub-models drive the estimate, and what the alternative explanations are.⁴ The market's 72% comes with *timeliness*: biosecurity experts who have examined the specific outputs, researchers who have attempted similar experiments, insiders with evaluation data the published literature does not yet contain.⁵ She needs both. So she tries to reconcile them.

But the market's 72% is not one signal. It combines genuinely independent evidence (insider data, reproduction attempts), AMTAIR-derived reasoning (forecasters citing the five-hypothesis decomposition), and shared-literature correlation (common priors from overlapping source material).⁶ The genuine market

¹This vignette is hypothetical. AMTAIR currently exists as a prototype and thesis project (Section I.2); the prediction market integration, institutional scaling, and organizational dynamics analyzed in this paper are prospective. The scenario illustrates the problem space the paper's methodology addresses.

²If the model provides genuinely novel synthesis guidance beyond publicly accessible sources, emergency deployment restriction and biosecurity agency notification are warranted. If it aggregates publicly available information in a more accessible format without adding new capability, a dual-use research policy review suffices — but no emergency restriction. An evaluation elicitation artifact (specific prompt strategies producing alarming-looking but non-actionable outputs) warrants methodology revision. Training data memorization of dual-use research papers warrants a data audit. Scaffolding-contingent capability (requiring agentic tool access a lone actor would not have) warrants scaffolding governance, not model restriction.

³The dashboard shows posteriors conditioned on the leaked evaluation evidence: de novo synthesis capability 35% (practically, AMTAIR would deliver the full probability distributions across the models used, but for summarization the mean and the 90% confidence interval, i.e., (20%, 52%), are useful summary statistics), recombinative accessibility 28% (15%, 43%), evaluation elicitation artifact 20% (8%, 34%), training data memorization 12% (4%, 24%), scaffolding-contingent capability 5% (1%, 14%) (all 90% confidence intervals). The ensemble aggregates several hundred quality-weighted extracted models — Bayesian networks formalized from published biosecurity and AI capability research, including dual-use risk frameworks and capability-threshold analyses (see Appendix C for a worked example). Extraction quality weights w_i can be derived from cross-validated extraction accuracy and source-argument independence scores. (Note that these model-level extraction weights w_i are distinct from the forecaster-level quality parameters α_i introduced in Section II.5, which measure forecaster reliability via Brier scores and calibration.)

⁴AMTAIR's Bayesian networks could trace each hypothesis through capability preconditions, training data provenance, and evaluation methodology. This auditable reasoning chain is AMTAIR's core strength and what the brief requires: policymakers need to understand *why* the agency recommends a particular response and what would change that recommendation.

⁵Academic publication timelines create a structural lag: peer review, revision cycles, and editorial processing typically span 12–24 months from submission to print. AMTAIR's extracted models inherit this lag; a Bayesian network formalized from a 2024 dual-use risk assessment may encode beliefs formed before the model in question existed.

⁶Three structurally distinct information channels compose the market signal (the decomposition is approximate — individual forecasters' reasoning does not divide cleanly into independent and contaminated components; the categories identify structurally distinct information channels, not measurable fractions of any individual's belief). Genuinely new information — insider evaluation data, reproduction attempts, expert biosecurity assessments — is the market's distinctive contribution. AMTAIR-derived reasoning — forecasters directly citing AMTAIR's decomposition, using its categories to interpret results — means the framework designed to structure assessment has become the frame through which the market interprets the incident. Shared-literature correlation — forecasters reasoning from the same published research AMTAIR extracted from — creates common-cause dependency even between forecasters who never saw AMTAIR's output. The common-cause structure runs deep: AMTAIR's models and the forecasters' reasoning both draw on overlapping literatures — dual-use risk frameworks, capability-threshold analyses, biosecurity evaluation methodology; a forecaster who reads Carlsmith (2022) and a Bayesian network extracted from Carlsmith (2022) share information even if neither is aware of the other. These three information channels (content composition of the market signal) cut across the three contamination

signal — the evidence she most needs — is trapped inside this composite. She cannot simply extract the market's timeliness from the contaminated whole.⁷

I.2 AMTAIR and Its Constitutive Tension

AMTAIR (developed as my thesis project) uses large language models to extract Bayesian networks from natural language AI safety arguments through a two-stage pipeline (ArgDown extraction followed by BayesDown enrichment), targeting comprehensive *libraries* of formalized risk models covering the field's major arguments.⁸ The system is designed as shared epistemic infrastructure: publicly accessible through open-source repositories with dashboard interfaces, community-governed through an author refinement process, and targeting broad adoption across AI safety organizations. Each of these properties simultaneously creates value and opens a contamination channel which if unchecked can result in information cascades. Section II.1 examines the specific mechanisms; the essential point for the thesis is that the duality is constitutive, not incidental.

The tension is constitutive: each functional requirement generates a contamination channel. Cross-stakeholder coordination requires publicity, but publicity exposes outputs to forecasters; calibration against diverse perspectives requires broad uptake, but uptake correlates beliefs; and policy relevance requires institutional authority from transparent methodology, but authority causes updating. A private, obscure, unused system would avoid contamination — but it would also be pointless. This paper refers to this structural duality — where value-creating properties and contamination-creating properties are the same properties — as the *constitutive tension*.

This creates a five-step feedback loop: AMTAIR publishes estimates; forecasters observe and update toward them; market prices shift; AMTAIR ingests the contaminated prices as ostensibly "independent" evidence; model confidence increases or decreases spuriously.⁹ This is worse than either source alone: a static model at least preserves its original information content cleanly; uncontaminated market prices at least reflect genuine independent beliefs. The feedback loop could corrupt both.

I.3 Research Question and Thesis

Research Question: How can AMTAIR integrate prediction market probabilities while managing the circular contamination risks that its design as shared epistemic infrastructure constitutively creates?

Thesis: AMTAIR's contamination problem is constitutive of its design: properties the system needs to fulfill its purpose (publicity, authority, broad uptake) are also properties that create contamination risk. Powerful statistical tools (hierarchical Bayesian frameworks, randomized embargoes, quality-weighted aggregation) exist for this problem, but the most effective techniques require full platform control that AMTAIR's open-loop architecture does not provide, and even full control leaves framing contamination intact. Under AMTAIR's actual constraints, practical statistical tools still help but each requires

mechanisms developed in Section II.5 (one-shot anchoring, feedback loops, framing — which describe how contamination propagates statistically). See Appendix A.6 for the analyst's practical workflow integrating both perspectives.

⁷If she feeds the headline 72% back into the ensemble, the mean drifts toward a price that already contains AMTAIR's own signal; this and successive cycles compress genuine disagreement into spurious convergence. She is not comparing two independent assessments but navigating one epistemic ecosystem from two angles. Appendix A.6 returns to this analyst's scenario and traces, step by step, what the formal apparatus (A.1-A.5) would prescribe, what she could do under idealized conditions, and what she can actually do given AMTAIR's open source but dashboard-only architecture — making concrete where each organizational dependency enters.

⁸See Appendix C for a worked example. The companion paper (V. J. Meyer, 2025d) develops AMTAIR's design rationale in detail: the ArgDown → BayesDown pipeline, multi-functional queryability following Veit (2019)'s strong model pluralism and the E/P/I framework, cost reduction from manual formalization, and the "level-separation" principle.

⁹The mechanism at the core of this feedback loop is *uncertainty laundering*: the model's original uncertainty gets obscured through circular feedback, returning as false validation — variance collapses not because genuine evidence accumulates, but because the same signal recirculates. The term "laundering" here differs from Gelman (2015)'s technical usage; we use it to describe how uncertainty about model validity gets obscured through circular feedback. Section II.5 classifies this as *feedback loop contamination*, with uncertainty laundering as its characteristic effect.

organizational infrastructure it does not generate: track-record systems, exposure metadata, institutional commitment to embargo enforcement. Organizational epistemology, specifically the analysis of epistemic vices at the collective level, identifies what infrastructure is needed and explains why it will not emerge spontaneously — each vice maps a specific mechanism through which organizational dynamics prevent the creation of a statistical prerequisite¹⁰. The resulting organizational mitigation, while imperfect, makes managed integration categorically preferable to the alternative of unmanaged informal integration.

Why integrate at all, given these risks? Because separation merely accepts the staleness problem, risks *unidentified* information cascades and forgoes substantial benefits (Section II.3). More fundamentally, practitioners will connect these sources informally regardless; the question is whether integration is managed rigorously or left to ad hoc practice.

I.4 Roadmap

The paper proceeds through four connected questions, each building on the previous section's limitations. *Why integrate* despite these risks? Section II argues that the staleness problem makes integration necessary, identifies the structural benefits, and examines the statistical response — showing that powerful tools exist under full platform control but fall short under AMTAIR's open-loop architecture, and that even full control leaves framing contamination intact. *Why does this generate an organizational problem?* Section III shows that the infrastructure practical statistics requires is unlikely to emerge spontaneously: epistemic vices, at both the individual and collective level, systematically inhibit the creation and maintenance of the organizational prerequisites the statistical framework presupposes. *How can the problem be approached?* Section IV develops organizational responses derived from the vice → vulnerability mapping: design requirements motivated by the specific mechanisms through which each vice prevents a statistical prerequisite from functioning. *Is the problem fully solved?* No; but managed integration, with named vulnerabilities and organizational infrastructure designed to maintain the conditions statistical tools require, is categorically preferable to unmanaged informal integration.

II. Integration and Its Costs: From Design to Contamination

II.1 AMTAIR's Design and Its Contamination Shadows

Each design feature that makes AMTAIR valuable simultaneously opens a contamination channel.

AMTAIR's two-stage pipeline (ArgDown extraction (Voigt, 2014) followed by BayesDown enrichment (V. J. Meyer, 2025a)) produces complete Bayesian networks from natural language arguments. LLM-automated extraction scales to comprehensive *libraries* covering the field's major arguments, but scale creates contamination surface: comprehensive libraries displayed on public dashboards can become shared reference frames from which contamination propagates.

Four design features illustrate the constitutive pattern: public accessibility, ensemble methods, author refinement, and broad uptake. They simultaneously create value and open contamination channels, becoming more consequential as the original data underlying the models becomes stale.¹¹

¹⁰The main statistical prerequisite is "independence". To understand the core intuitions behind these issues, imagine the attempt to isolate the counterfactual predictions, forecasts, analysis and so forth which may have been produced had neither AMTAIR nor any derivative works based on AMTAIR ever existed. While this might still be relatively easy to imagine at this stage, if AMTAIR becomes as widely used and successful as it aspires, these concerns would become exponentially less tractable. Conversely, imagine isolating all IPCC work from climate prediction models to capture where climate science would be if neither the IPCC nor any derivative work had existed.

¹¹Public accessibility exposes probability estimates to forecasters whose independent judgments the system needs. Ensemble methods partially mitigate anchoring on specific numbers but leave framing contamination intact — the deeper influence

II.2 The Staleness Problem

AMTAIR models extracted from published literature embed the epistemic state of authors at writing time. Academic publication cycles measured in years confront AI capability developments measured in months and weeks. A risk assessment written before a major capability demonstration cannot incorporate what that demonstration revealed. The pace is concrete: informal AI capability predictions from 2021 proved remarkably accurate regarding subsequent developments (Kokotajlo et al., 2025), with policy-relevant scenario analyses attracting attention within months. Models, predictions and data formalized from year-old sources are quickly outdated.

Two distinct staleness problems warrant distinction. *Parametric staleness* occurs when probability estimates become outdated but the underlying argumentative structure remains valid; the network topology is correct, only the numbers need updating. *Structural staleness* occurs when entire argumentative frameworks become obsolete: a model updating 2019 parameters¹² of a compute growth model while the field debates 2025 concepts operates on an obsolete ontology. The integration approach developed here — incorporating prediction market data into AMTAIR's extracted models — addresses parametric staleness directly by providing continuously updated probability estimates; structural staleness requires more fundamental model revision.

In AMTAIR's ensemble architecture, correlated staleness compounds both failure modes — shared empirical sources create correlated parameter outdatedness, shared theoretical frameworks create correlated structural obsolescence — making external integration a design-level necessity, not an option.¹³

Prediction markets offer a bridge. Markets respond to new developments within hours or minutes, not months, providing continuously updated probability estimates and information processing from distributed intelligence. But this bridge comes at a cost, one determined by the specific design of the system being integrated.

II.3 Why Integrate Despite the Risk?

Given these costs, why pursue integration at all? The alternative — leaving AMTAIR's extracted models and prediction market forecasts as unconnected epistemic resources — accepts the staleness problem, creates work for users, makes AMTAIR significantly less useful for policy purposes and forgoes structural benefits specific to AMTAIR's architecture, not generic properties of prediction markets.

Prediction market integration reconfigures the epistemic incentives and environment in which beliefs are formed, expressed, and aggregated — a deeper contribution than updated numbers. What de Bruin (2013) calls the operationalization of epistemic virtues at the structural level — designing institutions so that the conditions for good epistemic conduct are built into organizational architecture rather than left to individual disposition — is precisely what integration achieves. Two dimensions of this reconfiguration are specific to AMTAIR.

channel of shared causal decompositions remains untouched (Section II.5). Author refinement risks within-repository contamination when authors adjust formalizations to match priors, requiring organizational governance of the refinement process. Broad uptake correlates forecasters' beliefs with AMTAIR's models even without direct consultation, extending the contamination surface beyond direct dashboard visitors. Appendix C illustrates these channels concretely through the extraction of Carlsmith (2022).

¹²An important exception here occurs when models have "foreseen" certain events or state of the world (e.g. a specific capability) in the sense that they included them in the model as nodes. In such cases, the probability of such nodes can be updated to 1 (e.g. if the event occurred) and the model may continue to function well for other purposes.

¹³Without external information to challenge shared assumptions, correlated staleness is self-reinforcing — the ensemble converges not because the models agree on the truth, but because they share the same blind spots. Entire libraries of extracted models may embed outdated assumptions; if the field undergoes a paradigm shift, the shared framing across multiple formalizations reflects a superseded understanding. But the channels through which external information enters are also channels through which contamination propagates (Section III).

Structural transparency through Bayesian network decomposition reveals which conditional dependencies drive conclusions. When market prices diverge from model outputs, the BN structure tells you where to look for disagreements.¹⁴ Cross-boundary aggregation for a fragmented field creates numerically commensurable outputs from disparate communities — a shared epistemic currency that can bypass organizational boundaries.¹⁵

Each benefit creates a corresponding contamination channel (transparency exposes outputs to forecasters, cross-boundary aggregation correlates beliefs across communities, accountability incentivizes attending to AMTAIR's authoritative outputs), which Section III analyzes.

II.4 DAGGRE: What Closed-Loop Systems Avoid

Bayesian network–forecast integration is not hypothetical. IARPA's DAGGRE project demonstrated that it works and its architecture also reveals where and why AMTAIR's situation differs. DAGGRE showed that decomposing complex forecasts through Bayesian network structure improves accuracy and that BN-based automated trading algorithms could exploit expert-specified causal structure.¹⁶

DAGGRE was a closed-loop system: feedback stayed internal within a single IARPA competition platform. AMTAIR's aspiration to function as shared epistemic infrastructure requires open-loop bidirectionality with external forecasting communities across multiple platforms (see Figure 1).¹⁷ This structural difference — closed-loop containment versus open-loop exposure — determines whether information silos are feasible or if the contamination problem organizationally and structurally embedded.

¹⁴This diagnostic specificity directly counters organizational complexity (M. Meyer, 2023): opaque reasoning becomes auditable at the level of individual inferential steps. Epistemic hubris (Baird & Calvard, 2019) becomes visible through calibration tracking. This is not a benefit forecasting platforms offer by default; it is a benefit of integrating prediction markets with structured causal models.

¹⁵The AI safety research community is fragmented across academic labs, non-profit organizations, frontier AI companies, advocacy groups, and government agencies — each with different epistemic cultures, different access to information, and different incentive structures. This is fragmentation (M. Meyer, 2023) at the inter-organizational level. Integration incorporates aggregated judgment from thousands of forecasters, many with relevant expertise who never publish academic papers; Tetlock & Gardner (2015) demonstrate that such aggregate judgment consistently outperforms individual experts. Markets counter epistemic insouciance (Baird & Calvard, 2019): financial stakes and public track records create accountability absent from traditional expert elicitation (Frankfurt, 2005). Track-record-based weighting structurally bypasses epistemic injustice (Baird & Calvard, 2019): a well-calibrated junior analyst receives higher weight than a senior researcher with poor track record.

¹⁶Key DAGGRE findings: combinatorial conditional forecasts provide information unavailable from standard single-question markets; BN-enforced consistency constraints prevent common probabilistic errors; and BN-based autotradars could identify arbitrage opportunities in forecast data (Berea et al., 2012; Berea & Twardy, 2013; Powell et al., 2013). Important caveats: Powell et al. (2013) found mixed results for combinatorial markets versus flat markets, and DAGGRE's combinatorial feature saw limited adoption. AMTAIR's approach differs — integrating existing unconditional forecasts post-hoc rather than requiring forecasters to trade on conditionals directly. AMTAIR's LLM-assisted extraction automates similar structural specifications that DAGGRE's autotradars exploited.

¹⁷AMTAIR publishes extracted models on its own dashboard while also open sourcing all of its methodology, tools, code, infrastructure and corpus of work; forecasters who encounter these outputs carry framings and probability anchors to Metaculus, Manifold, Kalshi, Polymarket and elsewhere; AMTAIR then pulls forecast data from these platforms via their public APIs — an indirect contamination channel, mediated by human behavior, that DAGGRE's closed architecture never faced. DAGGRE was not shared public infrastructure; it was one tool among many, impressive but not widely referenced outside the intelligence forecasting community.

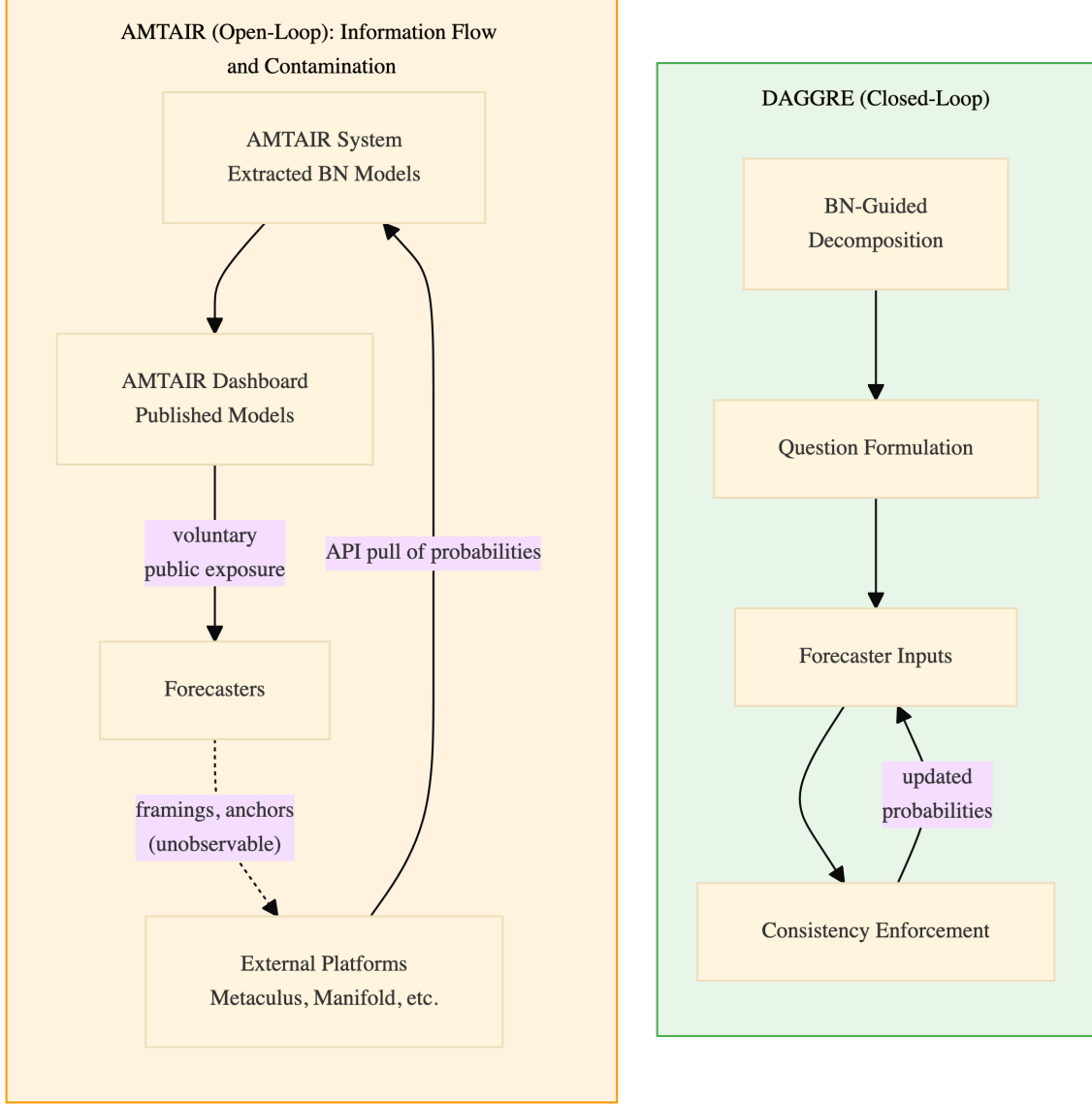


Figure 1: DAGGRE vs AMTAIR Information Flow: DAGGRE's closed-loop architecture contained feedback within a single platform. AMTAIR's open-loop design publishes to its own dashboard; forecasters carry framings and anchors to external platforms; AMTAIR then pulls forecast data via APIs — creating an indirect contamination channel mediated by human behavior.

DAGGRE could avoid contamination because DAGGRE was not trying to be what AMTAIR must be. The question is not whether PM-BN integration works (DAGGRE demonstrated that it can) but how to manage the contamination that authoritative public systems generate as a byproduct of their authority and usage.

II.5 A Naive Statistical Response

The obvious response to the contamination problem is statistical. When forecasters observe AMTAIR's published outputs, they anchor on a shared signal; naively recombining their forecasts with the model that produced that signal double-counts evidence.¹⁸

¹⁸The problem can be stated precisely: we want to estimate $P(\theta \mid M, F)$ — the probability of some parameter θ given both model outputs M and forecasts F — but standard Bayesian updating assumes $P(F_i \mid \theta, M) = P(F_i \mid \theta)$, i.e., that forecasts are conditionally independent of model outputs given the true state. When forecasters observe AMTAIR's outputs, this independence assumption breaks. If AMTAIR publishes $P(\text{catastrophic AI accident}) = 0.30$ and forecasters shift toward

In principle a hierarchical Bayesian framework addresses this by modeling each forecaster as an imperfect sensor of reality, characterized by quality (α_i) and contamination exposure (γ_i).¹⁹

Under the full platform control that DAGGRE enjoyed (Section II.4), these tools become substantially more powerful — contamination becomes not merely detectable but experimentally measurable.²⁰

But even under full platform control, one contamination channel persists. AMTAIR's causal decompositions must be public for peer review, and these structure how forecasters reason, creating framing contamination that resists randomization of numerical outputs. This principled limitation means the organizational gap exists regardless of platform control, not only because of AMTAIR's practical constraints.

Under AMTAIR's actual dashboard-only architecture, these tools degrade but remain useful. But each practical tool requires an organizational input it does not generate: quality parameters (α_i) require track-record infrastructure maintained by platform organizations; contamination parameters (γ_i) require exposure metadata that someone must design and fund; the critical quantity K — the number of forecasters who observed AMTAIR outputs — becomes unobservable without organizational tracking; embargo enforcement requires institutional commitment, not software configuration.²¹

Three distinct contamination mechanisms underlie this gap: one-shot probabilistic anchoring (mean bias from a single exposure), feedback loop contamination (variance collapse through multi-round uncertainty laundering), and framing contamination (dimensionality reduction from shared causal decompositions).²²

The gap between statistical prescription and AMTAIR's reality has two components, both organizational. The *practical* component: AMTAIR's architecture does not provide the conditions for full statistical control, and the practical tools that remain require organizational inputs they do not generate. The *principled* component: even under full platform control, framing contamination persists through the

0.30 after observing this, re-incorporating their estimates as independent information overcounts the 0.30 signal. The more such channels the worse the errors from confounding. See Appendix A.1–A.2 for formal specification.

¹⁹Rather than treating forecasts as equivalent independent samples, the hierarchy models the generating process behind each one — separating genuine private information from reflected AMTAIR signal — and downweights the contaminated portion. This addresses forecast-level independence violations; a distinct but related problem is model-level correlation, addressed in the companion paper (V. J. Meyer, 2025d) through Bayesian Model Averaging. Three tools operationalize this: the hierarchical framework itself, randomized temporal embargoes, and quality-weighted multi-platform aggregation. Embargoes withhold AMTAIR outputs during randomly selected periods, creating uncontaminated comparison groups. Quality-weighted aggregation gives better-calibrated forecasters more influence. See Appendix A.1–A.5 for formal specification (A.1–A.2 develop the quality-adjusted framework; A.5 develops the contamination correction).

²⁰Randomized assignment of which forecasters see AMTAIR outputs makes the contamination parameter γ_i estimable per forecaster. Temporal embargoes with known exposure groups enable clean difference-in-differences identification (consider a controlled setup where forecasts with and without AMTAIR exposure can be compared — such designs would enable causal, quantitative estimates of contamination effect sizes). The effective sample size, reduced from nominal N by correlated anchoring among K exposed forecasters to $N_{\text{eff}} = (N - K) + K/(1 + (K - 1)\rho)$, is recoverable because K is known by construction. Appendix A.5 develops five distinct randomization techniques and their formal requirements under each control level.

²¹Population-level embargo comparisons substitute for individual randomization, quality weighting draws on platform-maintained scores, and multi-platform aggregation exploits cross-platform variation. Individual-level randomization captures only direct dashboard visitors; a single blog post summarizing outputs contaminates forecasters who never visited. A further complication: some platforms provide only aggregate market prices (Polymarket), not individual forecasts (Metaculus) — which platforms AMTAIR integrates with is itself an organizational design choice with direct statistical consequences. Appendix A.5 develops this formally. Alternative statistical approaches — finite mixture models, instrumental variables, propensity score matching, copula models, causal discovery algorithms — can substitute different assumptions for different organizational inputs, but none eliminates the organizational dependency; see A.6.

²²The first two are substantially addressable under full-control conditions and partially mitigable under AMTAIR's constraints — by randomization and embargo mechanisms respectively. The third is partially detectable (via framing contamination metrics; see A.4) and partially mitigable (via alternative-decomposition elicitation and diversity incentives), but not eliminable, because the argument structures that constitute the framing channel must remain public for peer review. Section III.1 reclassifies these three mechanisms by the organizational infrastructure each requires. On the three types in detail: (1) One-shot probabilistic contamination requires no feedback loop — a single exposure produces mean bias, addressable by randomization. (2) Feedback loop contamination: AMTAIR publishes $\rightarrow$ forecasters anchor $\rightarrow$ market prices shift $\rightarrow$ AMTAIR re-ingests with tighter confidence $\rightarrow$ next round of stronger anchoring. Produces variance collapse through uncertainty laundering, addressable by embargo. (3) Framing contamination requires no feedback loop and no numerical exposure. Produces dimensionality reduction. Constitutive of AMTAIR's public infrastructure requirement. Appendix A.5 decomposes these into five contamination signatures (i)–(v) and maps each to organizational requirements under full-control vs. dashboard-only conditions.

causal decompositions that must remain public.²³

Section III examines why organizational infrastructure that would reduce both components of this gap will not emerge spontaneously — and how vices that prevent it are causal explanations of institutional failure.

III. The Organizational Nature of Contamination: Vices as Vulnerabilities

III.1 The Identification Problem and Its Organizational Resolution

Section II established that even practical statistical tools require organizational inputs they do not generate. This section examines why those inputs will not appear spontaneously — and why the gap is not a temporary practical limitation but a structural feature of the relationship between statistical frameworks and organizational information environments.

The hierarchical Bayesian framework outlined in Section II requires estimating two parameters for each forecast: quality (α_i) and contamination level (γ_i). For well-resolved questions — those that eventually come true or false — track records can approximate α_i well. But AMTAIR addresses long-horizon questions about transformative AI that may not resolve for decades. Without access to a reliable approximation of ground truth, there is no objective signal to distinguish a high-quality forecaster making a legitimate update from a "contaminated" forecaster just parroting model outputs. Track records from resolved questions assume cross-domain stationarity in forecaster competence but a forecaster well-calibrated on near-term geopolitics may lack relevant expertise for novel and out-of-distribution AI capability questions.

What fills this gap? The statistical framework depends on three categories of organizational facts it does not produce. These correspond to the three dimensions through which Choo (2008, pp. 32–34) analyzes organizational information processing — information needs, information seeking, and information use — providing analytical vocabulary for what would otherwise be an ad hoc list of dependencies.

Quality parameters (α_i) are not easily observable properties of individuals but require records maintained by platform organizations — and without organizational commitment to tracking, α_i remains unobservable.²⁴

The contamination parameter introduces a more consequential organizational dependence: to which degree a forecaster is influenced by AMTAIR outputs depends on organizational information flows and the entire ecosystem — publication timing, discussion threads, workshop presentations — estimates that the statistical framework relies on but does not generate or control.²⁵

Contamination also operates through shared *framing* — how AMTAIR and the underlying models decompose problems into causal structures is itself a contamination channel, risking not only the distortion of estimates but even of the *ontology* underlying the field.²⁶

²³Contamination also operates through channels the statistical framework does not model — authority relations, hierarchical deference, and structural impediments to information flow — which determine whether statistical prescriptions are actually followed.

²⁴Metaculus tracks Brier scores; Manifold computes calibration curves; Good Judgment certifies superforecasters. These track records exist because organizations invest in measurement infrastructure — unglamorous work involving database maintenance, resolution criteria adjudication, and longitudinal record-keeping. This corresponds to Choo's (2008) information use dimension.

²⁵Did the forecaster read the dashboard? Follow a discussion thread referencing AMTAIR probabilities? Attend a workshop where AMTAIR decompositions were presented? (Even where AMTAIR aims to purely represent models differently, spurious confirmation can still arise — simply presenting information differently can bias subsequent forecasts.) These are facts about organizational communication channels, not statistical quantities. Publication timing — an organizational decision — determines which forecasts were collected under embargo. This corresponds to Choo's (2008) information seeking dimension.

²⁶With multiple extractions of the same arguments, the relevant question is not just *did the forecaster see AMTAIR's output?* but *which extraction did they encounter?* Even where AMTAIR's representations appear clearly correct, the

This analysis identifies three organizational gaps — quality-parameter estimation (α), exposure tracking (γ), and ontological framing — which cut across the contamination mechanisms distinguished in Section II.5. But the taxonomy does not demonstrate exhaustiveness: further channels — for example *temporal contamination*, where AMTAIR's outputs shape the research agenda that produces future extracted literature — illustrate that the identified mechanisms are salient exemplars, not jointly exhaustive categories.²⁷

The identification problem is therefore not a mathematical gap awaiting a better estimator. It is a gap that exists *because* the relevant parameters arise from organizational information environments *outside* the reach of the formal apparatus. Section III.2 explains why the organizational infrastructure that practical statistical tools require will not emerge spontaneously, and why the vices analyzed there are not derogative labels but causal explanations of institutional failure.

III.2 Organizational Amplification: Why Safeguards Fail

Section III.1 established that the statistical framework does not *know* what it needs to know: its parameters require organizational infrastructure. But suppose that problem were solved: all parameters perfectly estimated, all track records maintained, all exposure metadata collected and so forth. The safeguards prescribed would still be difficult to sustain, because organizational dynamics intervene between prescription and implementation. This is the amplification problem: knowing the right answer is not the same as being able to act on it when institutional pressures push in the opposite direction.

The vices analyzed below are not labels applied after the fact but causal explanations of why the infrastructure that statistical tools need is unlikely to emerge spontaneously; each vice identifies a specific mechanism through which organizational dynamics impede the creation or maintenance of a statistical prerequisite.

Information cascades (Bikhchandani et al., 1992, pp. 992–996) — where rational agents discount private information in favor of observed public signals — are well-documented in prediction markets²⁸. Bikhchandani et al. (1992, pp. 994–995) identify three conditions for cascade formation: sequential observation of choices, limited private information, and rational Bayesian updating. AMTAIR satisfies all three: forecasters observe published model outputs (sequential observation), hold private information of varying quality about AI risk, and rationally update on the accumulated public signal AMTAIR provides. Almaatouq et al. (2020) show that network structure determines whether social information helps or hurts collective accuracy — adaptive networks (where connections depend on past accuracy) outperform static networks, directly supporting quality weighting mechanisms.

Prediction markets may contain built-in anti-cascade mechanisms, but AMTAIR's constitutive tension choice of which models to extract can still introduce systematic bias. This third dependency corresponds to Choo's (2008) information needs dimension: without a decomposition that distinguishes ontological influence from parametric influence, one might treat contamination as a single phenomenon requiring a single response. The three-dimensional structure identifies three distinct failure modes, each requiring different organizational infrastructure — measurement systems for α_i , communication tracking for γ_i , and institutional self-awareness about the categories the system imposes on its environment. Backtesting — training models using only information available at past time t , then comparing to subsequent developments — can partially validate the methodology on resolved questions, but for long-horizon AI governance questions, ground truth may remain permanently unavailable; this connects to Brundage et al. (2020)'s broader concern with supporting verifiable claims in AI development.

²⁷Temporal contamination differs from the ontological channel because the mechanism is not cognitive (forecasters internalizing framings) but institutional (funding bodies, journal editors, and research groups redirecting effort toward AMTAIR-salient questions), and the organizational response would accordingly differ — research-agenda auditing rather than forecaster-level contamination tracking. The three organizational gaps cut across the three contamination types: one-shot probabilistic, feedback loop, and framing contamination operate through different combinations of quality-parameter, exposure-tracking, and framing dependencies. The organizational responses developed in Section IV address the identified channels; they do not seal the system against all possible contamination.

²⁸See Appendix B.1 for an extended review of information cascade theory in prediction markets, including conditions under which cascades form and mechanisms through which they break.

could undermine them. If the anchoring signal originates from AMTAIR (outside infrastructure) and the markets are designed to feed back into AMTAIR, this can produce a self-enforcing feedback loop or information cascade.²⁹

Beyond cascades, AMTAIR faces organizational amplification mechanisms that cascade theory does not address. The epistemic vices identified by Baird & Calvard (2019) and M. Meyer (2023) explain *specific failure modes* as causal accounts of why safeguards break down.

Epistemic hubris explains why embargo protocols get overridden — AMTAIR's quantitative precision can create an illusion of certainty that makes contamination controls appear unnecessary.³⁰

Epistemic insouciance explains why exposure metadata may never get collected — AMTAIR's institutional setting as an under-resourced academic project places it outside compliance mandates, and contamination damage is subtle and delayed.³¹

Epistemic injustice explains why authority often overrides calibration — quantitative reasoning carries disproportionate epistemic credence in AMTAIR's operating communities, making disagreement with a formal system's outputs demand a higher burden of justification than optimal Bayesian frameworks prescribe.³²

The 2022 FTX forecasting failure illustrates how these vices can interact.³³

The above vices — hubris, insouciance, injustice — are individual dispositions *amplified* by organizational structures: particular people override embargoes, neglect metadata, defer to prestige. In principle, individual-level interventions could address them — training, incentive redesign, accountability mechanisms targeting specific actors. But M. Meyer (2023)'s empirical taxonomy — derived from analysis of 1.3 million employee reviews of 868 US corporations — identifies two genuinely collective vices irreducible to individual-level analysis.³⁴ The distinction shows what solutions are possible: even if no individual in

²⁹Financial incentives reward contrarian positions, price discovery reveals aggregate information, and Bikhchandani et al. (1992, p. 1003) note that cascades are fragile — a sufficiently informative public signal can break them. But these mechanisms assume the cascade's public signal is external to the market. When the crowd's prior and the model's output share a common cause, market incentives to disagree do not help — the anti-cascade mechanism (independent private information) is precisely what contamination erodes.

³⁰Epistemic hubris: overconfidence in one's epistemic position (Baird & Calvard, 2019). Bayesian networks output precise numerical probabilities ($P = 0.338$, not "somewhat likely"), and quantitative precision creates a greater illusion of certainty than verbal risk assessments. LLM-automated extraction compounds this — producing BNs dramatically faster than manual approaches, breeding the impression that outputs are easily verified. An institution confident in its extraction quality may not see the need for contamination controls. This amounts to what Choo (2008) calls risk denial.

³¹Epistemic insouciance: indifference toward accuracy (Baird & Calvard, 2019; cf. Frankfurt, 2005). AMTAIR operates outside the compliance mandates that force financial institutions to maintain audit trails; its developers are researchers optimizing for extraction quality, not data governance professionals; contamination damage is discoverable only when validation is attempted, possibly years later. Tracking which forecasters saw which outputs is costly, unglamorous, and has no immediately visible utility. The statistical models need γ_i , but insouciant institutions see little reason to invest in generating it. The author refinement loop illustrates this: authors updating extractions interact with probability outputs, but distinguishing genuine correction from prior-matching requires organizational governance — peer review, version tracking, justification requirements — that volunteer-based management might not maintain.

³²Epistemic injustice: discounting knowers based on status rather than competence (Baird & Calvard, 2019). The AI safety and Rationalist/EA forecasting communities arguably accord Bayesian reasoning greater cultural authority than do political science or epidemiology [author's observation; Yudkowsky (2015) establishes Bayesian epistemology as the rationalist community's central framework, but formal comparative sociology of epistemic norms across forecasting communities remains sparse]. A forecaster who disagrees with AMTAIR's $P = 0.338$ is implicitly disagreeing with transparent analytical reasoning. Quality weighting exists mathematically, but organizational prestige dynamics can override it: a well-calibrated junior analyst's forecast often carries less organizational weight than a formal system's output.

³³Metaculus question #7230 ("Will FTX default...?") stood at ~2% before resolving positively in November 2022 — among the platform's most miscalibrated high-profile predictions. Three mechanisms operated simultaneously: information cascades, where each low estimate made subsequent private concerns less credible (Bikhchandani et al., 1992, pp. 994–995); reputation-based deference, where Sam Bankman-Fried's status within the Effective Altruism community operated as social proof; and hidden private information inaccessible to markets. The interaction is what matters: a forecaster with private concerns saw both that the market price was low and that high-status figures weren't objecting — either signal alone might be overridden, but together they suppressed dissent.

³⁴Meyer identifies six corporate epistemic vices via factor analysis: incompetence, dishonesty, unreliability, fragmentation, ambiguity, and complexity. He classifies fragmentation and complexity as "genuinely" collective, drawing on Ridder (2022)'s three-way framework distinguishing additive, interactionist, and emergent collective vices. Meyer's empirical validation is scoped to single corporations; applying fragmentation and complexity to a multi-organizational ecosystem like AMTAIR's is this paper's extension, motivated by structural parallels: the governance fragmentation and process opacity that Meyer

the AMTAIR ecosystem is negligent or overconfident, the organization as a whole may fail to see and prevent contamination loops.

These are emergent properties of organizational structure itself because individual accountability mechanisms, however well designed, do not address what no individual is responsible for. They require structural redesign:

Fragmentation (organizational structures undermining knowledge sharing) means no single actor sees the whole contamination loop. The feedback spans AMTAIR operators, forecasting platforms, and consuming researchers and policy teams. The statistical model sees aggregates; few actors in the system see the organizational dynamics and aggregation algorithms producing those aggregates. This is not individual negligence but collective structural failure.

Complexity (complex or inefficient processes undermining organizational knowledge) means the feedback dynamics might be too intricate for any individual to trace. Multiple extractions, ensemble methods, the author refinement loop, cross-platform propagation — these create contamination pathways that an organization's own processes may render opaque to it.

III.3 The Analytical Summary: From Vices to Vulnerabilities

Having identified *which* gaps between statistical prescriptions and organizational realities are structural — the identification problem (III.1) and the amplification problem (III.2) — the next step is to operationalize the insight. The mapping below attempts this: each row identifies where a specific practical statistical tool depends on organizational infrastructure, which epistemic vice limits that infrastructure from appearing spontaneously, and what organizational responses could fill the gap:

Table 1: Vice→vulnerability mapping: Where the statistical solution requires organizational infrastructure.

Vice	AMTAIR Vulnerability	Why Statistics Does Not Address	Organizational Responses
Hubris (Baird & Calvard, 2019)	Embargo protocols overridden; outputs treated as certain	Framework assumes compliance with its own procedures	Commitment mechanisms; mandatory uncertainty foregrounding
Insouciance (Baird & Calvard, 2019)	No exposure metadata collected; extraction audit skipped	Framework needs γ_i but generates no incentive to collect it	Data governance protocols; incentivized tracking
Injustice (Baird & Calvard, 2019)	Institutional prestige overrides calibration track records	Quality weighting overridden by organizational hierarchy	Platform-level design; open-source contribution meritocracy
Fragmentation (M. Meyer, 2023)	Single actors at best observe only parts of the complete contamination loop	Models based on statistical aggregates, not organizational dynamics	Cross-organizational transparency; contamination auditing
Complexity (M. Meyer, 2023)	Feedback dynamics too opaque for any individual to trace	Requires organizational information flows model does not access	Version control; author refinement peer review

identifies within corporations characterize inter-organizational systems at least as strongly.

Each row maps organizational infrastructure the statistical framework would require but does not build for itself. Contamination is not merely a technical anomaly requiring a bespoke technical fix but a manifestation of well-characterized organizational epistemic failures, tractable through the same analytical framework that organizational epistemology has developed for other domains.

This mapping is productive beyond diagnosis: it can help to produce institution-specific prescriptions from general analytical categories.³⁵

The institution operating AMTAIR would itself be subject to every vice the table maps.³⁶ This is not infinite regress³⁷. Each governance layer reduces risk; none eliminates it. The point is structural: the managing organization is not external to the contamination dynamics — it operates *within* them.³⁸

IV. Toward Organizationally Embedded Integration: Institutional Design

Sections II and III developed a three-step elimination: powerful statistical tools exist for the contamination problem, but AMTAIR does not meet their conditions; practical tools still help, but require organizational infrastructure they do not generate; and organizational vices explain why that infrastructure is unlikely to emerge spontaneously. This section develops what the required infrastructure looks like, motivated by the vice → vulnerability mapping (Section III.3), which constrains the space of adequate responses without uniquely determining them.

These proposals must be understood as structured risk reduction *within* the system, not control imposed from outside it, because the organization managing contamination controls inhabits the same epistemic environment it seeks to govern (Section III.3). These interventions do not eliminate the tension; each reduces its severity while remaining vulnerable to the vices Section III mapped.

³⁵The vice taxonomies of Baird & Calvard (2019) and M. Meyer (2023), applied to AMTAIR's specific architecture, suggest the five-row mapping above; each row then motivates an organizational design requirement (Section IV) that the formal apparatus needs but does not build for itself. Mandatory uncertainty foregrounding, incentivized exposure tracking, cross-organizational auditing — these were not assumed as desiderata but motivated by the system-specific vulnerability analysis. The connection from vice to prescription involves judgment at each step rather than strict derivation.

³⁶An overconfident AMTAIR team may skip contamination checks (hubris). Resource-constrained management may neglect tracking infrastructure (insouciance). Internal hierarchies may override calibration data (injustice). No single team member may see the full contamination loop (fragmentation). The system's own processes may be opaque to those operating it (complexity).

³⁷AMTAIR's approach can mitigate the kind of process opacity that M. Meyer (2023) identifies as the vice of complexity in corporations. However, the AMTAIR team itself remains subject to the same epistemic challenge: its own processes may be too intricate for any single member to trace.

³⁸A deeper theoretical point: Milgrom & Stokey (1982) show that under common priors, rational Bayesian agents should not trade at all — prediction markets work precisely because participants hold genuinely different priors. But contamination erodes this diversity: as AMTAIR's framings permeate the forecasting community, priors converge, and the market's informational value diminishes — the dimensionality reduction mechanism formalized in Appendix A.5(iv). The no-trade theorem thus identifies the mechanism through which contamination is not merely a statistical nuisance but a structural threat to the market's epistemic function.

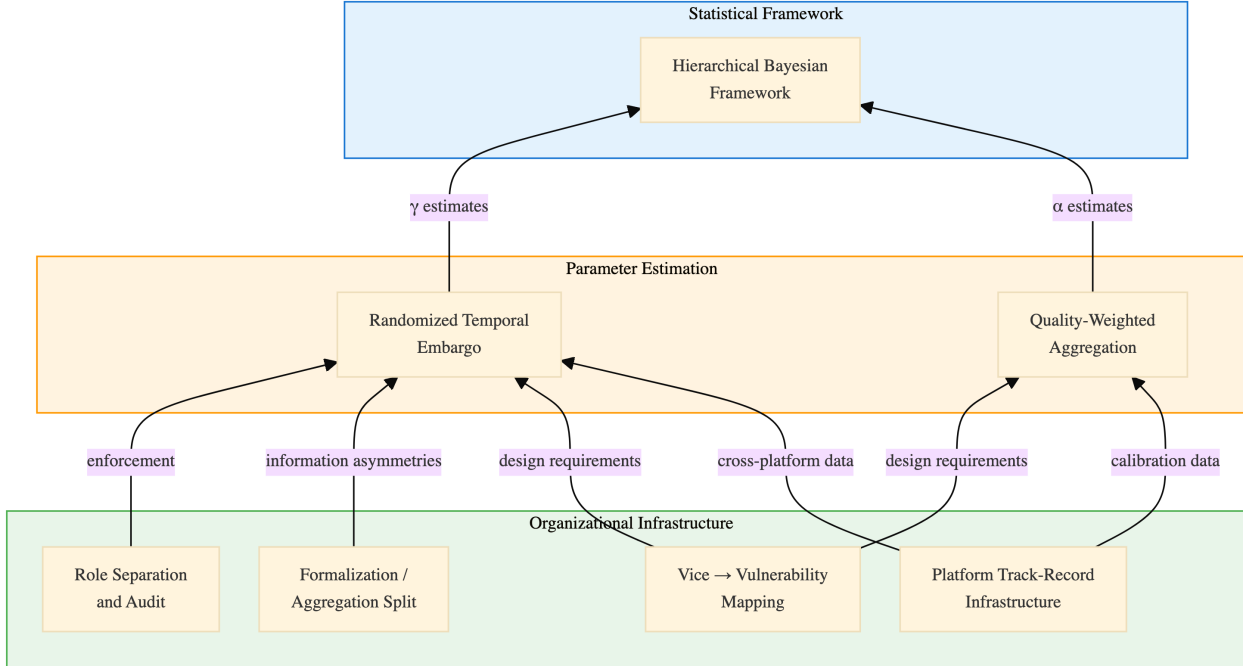


Figure 2: Integration Architecture: Statistical tools depend on organizational infrastructure. The hierarchical Bayesian framework (IV.1) requires quality parameters (α) maintained by platform track-record systems and contamination parameters (γ) derived from embargo data (IV.2). Both rest on institutional commitments: the vice→vulnerability mapping (III.3) motivates design requirements, role separation enables enforcement, the formalization/aggregation split creates information asymmetries, and platform track-record infrastructure both maintains calibration data for quality weighting and provides cross-platform forecaster identification needed for embargo estimation (comparing exposed vs. unexposed group means requires identifying individual forecasters). Arrows indicate dependency: upper layers depend on lower layers. Note: this diagram represents the parametric framework (contamination dimensions i–iii of A.5). Framing contamination (dimension iv) — requiring organizational culture rather than statistical infrastructure — is not represented; see A.5 for the full addressability gradient.

IV.1 The Hierarchical Framework in Organizational Practice

The hierarchical Bayesian framework remains the right mathematical structure: treating forecasts as noisy observations rather than direct evidence is the correct response to violated independence. What changes is the understanding of what can make the framework *work*. The required parameters are data about organizational information environments that someone must produce, maintain, and govern. This derives the organizational design specifications from the statistical prescriptions.³⁹

Quality parameters (α_i) depend on track-record infrastructure *maintained by platform organizations* — Metaculus Brier scores, Good Judgment superforecaster certifications, Manifold calibration curves. These examples exist because idealistic organizations invested in measurements. Contamination parameters (γ_i) depend on exposure metadata — *organizational publication decisions* and information flow documentation that someone must design, fund, and maintain. The framework's weights reflect *organizational risk tolerance* and value-laden choices about which ensemble methods to trust.⁴⁰

³⁹See Appendix A for a formal specification; A.6 traces the full practical workflow from an analyst's perspective, showing where each organizational dependency enters

⁴⁰This framework complements the Bayesian Model Averaging infrastructure developed in the companion paper (V. J. Meyer, 2025d): where BMA maintains weighted ensembles of extracted models, the framework here addresses what happens when external forecast data is ingested into those ensembles. Contamination distorts the full statistical signature through multiple channels — mean bias toward published values, variance underestimation through multi-round laundering, potential variance overestimation from conflicting ensemble models, and dimensionality reduction in reasoning — producing spurious convergence that mimics but does not constitute genuine evidence accumulation. See Appendix A.5 for formal treatment.

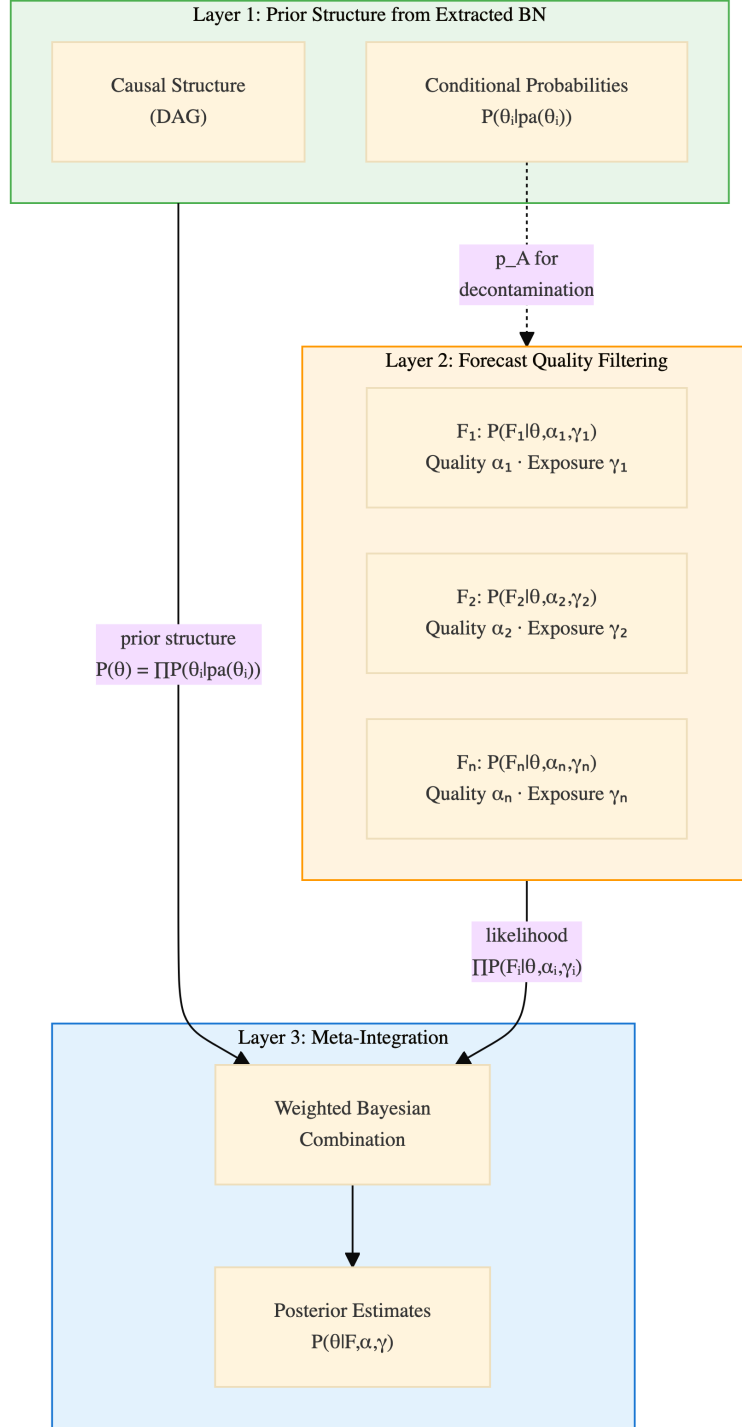


Figure 3: Three-Layer Hierarchical Bayesian Framework: Extracted Bayesian networks provide prior structure (Layer 1). External forecasts are weighted by organizationally maintained quality scores α and adjusted for contamination exposure γ (Layer 2). The contamination correction requires AMTAIR's published value p_A from Layer 1 (dashed arrow): decontaminating each forecast requires the same prior output that the forecast may have been contaminated by. Meta-integration combines both layers into quality-adjusted probability estimates (Layer 3). Each parameter depends on organizational infrastructure: α on platform track records, γ on embargo and exposure metadata. Note: this diagram represents the parametric contamination channels (dimensions i–iii of A.5). Framing contamination (dimension iv) — influence through shared causal decompositions rather than shared numbers — is not captured by the α/γ parameterization and requires organizational rather than statistical infrastructure (see A.5, dimension iv).

The vice $\rightarrow$ vulnerability mapping in Section III identified organizational conditions without which the Bayesian machinery would be unlikely to function reliably. The responses developed below illustrate how those conditions might be met. Each is analytically motivated — the mapping suggests lower bounds on what kind of organizational response is needed — but the specific prescriptions are examples of directionally helpful responses, not the only possible ones.

The individual/collective vice distinction shapes the space of adequate responses: where a vice is an individual disposition amplified by structures, interventions can target practices within existing arrangements; where a vice is genuinely collective, the organizational structures themselves must change.⁴¹

Three individual vices call for targeted interventions within existing arrangements. Against *hubris* (embargo override), mandatory uncertainty foregrounding — displaying confidence intervals and model disagreement prominently — can reduce the precision of the anchoring signal that creates false certainty, attenuating the contamination parameter γ_i the statistical framework must estimate.⁴² Against *insouciance* (metadata neglect), incentivized exposure tracking — embedding exposure logging as a default pipeline feature rather than relying on voluntary reporting — can make γ_i observable; without this organizational investment, the contamination correction apparatus operates on assumptions rather than data.⁴³ Against *injustice* (prestige over calibration), open-source contribution meritocracy can ground quality parameters α_i in demonstrated forecasting accuracy rather than institutional affiliation — a condition for the statistical framework's likelihood weighting to function as designed.⁴⁴

The two collective vices require structural redesign rather than individual-level intervention. Against *fragmentation* (nobody sees the loop), cross-organizational contamination auditing by structurally independent auditors with visibility across all three organizational boundaries addresses what no individual-level accountability mechanism can: the fact that the contamination loop spans organizational boundaries that no single participant oversees.⁴⁵ Against *complexity* (opaque processes), version-controlled extraction with mandatory peer review could create institutional records that makes the contamination chain reconstructable — and enables embargo analysis, since difference-in-differences estimation requires knowing exactly what was published when.⁴⁶

⁴¹Even this constraint leaves the specific institutional design substantially underdetermined — providing flexibility for adaptation to specific institutional settings.

⁴²Formally, hubris amplifies the variance collapse mechanism (A.5, dimension ii): overconfident engagement with point estimates strengthens γ_i , accelerating multi-round laundering in which exposed forecasters' variance decays at rate $(1 - \gamma)^2$ per feedback cycle. Communicating "P = 0.338 (0.15, 0.55), 3 models disagree" rather than "P = 0.338" makes the object that generates false certainty simultaneously communicate its own uncertainty — a wide interval anchors less strongly than a point estimate.

⁴³Exposure tracking is the organizational prerequisite for computing K (the number of exposed forecasters) in the effective sample size formula $N_{\text{eff}} = (N - K) + K/(1 + (K - 1)\rho)$, and for estimating individual γ_i (A.5). Automatic timestamp comparison and platform API integration would convert metadata collection from a discretionary cost into a default feature. See A.6 for the practical consequences of unobservable K .

⁴⁴This directly determines what α_i (A.1) measures: if quality weights reflect institutional prestige rather than demonstrated calibration, the hierarchical framework's likelihood weighting $P(F_i | \theta, \alpha_i)$ systematically overweights well-connected forecasters and underweights well-calibrated ones — embedding epistemic injustice in the statistical apparatus itself. Publicly verifiable metrics (extraction accuracy, calibration scores, version-controlled model contributions) make the basis for status-based discounting visible and contestable.

⁴⁵Concretely, designated auditors not embedded in any single component would trace published AMTAIR estimates through the forecasting ecosystem and back. The audit function must be structurally independent of the teams it monitors — reporting to governance oversight rather than to AMTAIR's development team, whose natural incentive is to demonstrate that the system works rather than to discover that it contaminates. This mirrors independent audit committees in corporate governance, where the auditor's institutional position (outside the operational hierarchy) is what makes the audit credible. Without structural independence, contamination auditing degenerates into self-assessment — the epistemic task that fragmentation structurally prevents. The cascade detection metrics (A.4) — temporal correlation, forecast clustering, cross-platform divergence decay, framing contamination index — provide measurement tools, but each requires data from multiple organizational boundaries that fragmentation makes invisible to any single participant.

⁴⁶Version control is the organizational precondition for temporal embargo analysis (A.5, technique 1): without a version-controlled record of which model versions, probability estimates, and structural decompositions were visible at each point in time, the embargo comparison has no baseline. When a contamination audit discovers anomalous forecast clustering around an AMTAIR estimate, the version history reveals which extractions were published when, which authors refined them, and what probability estimates were visible at each stage. Each modification to an extracted model would be documented,

Whatever specific form these responses take, they are organizational commitments, not technical features — they require institutional will, sustained funding, and deliberate cultural practices. Each can also be vulnerable to the vices it targets: mandatory uncertainty foregrounding can become a ritual that loses meaning (insouciance about the anti-hubris mechanism); contamination auditing can be captured by the very dynamics it monitors (the recursive problem). The goal is not organizational perfection but *named, monitored vulnerability* — categorically better than unnamed vulnerability, because named vulnerabilities can be tracked, measured, and improved upon even when they cannot be eliminated completely.⁴⁷

IV.2 Embargoes and Organizational Design

Randomized temporal embargoes provide the statistical leverage to estimate contamination effects by comparing forecast distributions with and without exposure.⁴⁸ But embargo design is fundamentally also an organizational question, not just a statistical one.

Pure embargo has limitations: AMTAIR outputs will eventually become public (indefinite embargo defeats the system's purpose), and in networked information environments, perfect information control is impractical. A natural response is organizational: separating the formalization layer (open, collaborative, allowing anyone to contribute to and inspect extracted models) from the aggregation layer (embargoed, with randomized sets of probability outputs withheld during randomly selected periods). This largely preserves AMTAIR's value as shared epistemic infrastructure while creating the informational asymmetries needed for contamination measurement.

Embargo enforcement is itself an organizational practice requiring commitment mechanisms. Choo (2008) identifies *structural impediment* as a key failure mode: warnings recognized by individuals may be blocked by organizational rules and hierarchies. Embargo protocols face the same risk: a junior analyst's observation that embargo was breached may never reach those responsible for data integrity. Clear escalation procedures, automated timestamp verification, and explicit organizational authority for embargo enforcement would transform embargo from a statistical mechanism into an organizational norm.

The formalization/aggregation separation illustrates this organizational design: maintaining open-source argument structures while withholding aggregated probability estimates at randomized intervals preserves collaborative benefits while creating the informational asymmetries contamination measurement requires.⁴⁹

The community overlap problem makes the organizational nature concrete — the people most qualified to operate AMTAIR are the same people who research and forecast in the field. One response is

justified against the source text, and reviewable by someone other than the modifying author. The organizational record substitutes for the individual cognition that complexity as a collective vice defeats.

⁴⁷Prediction markets themselves function as organizational architecture. Almaatouq et al. (2020) show that adaptive networks — where influence depends on demonstrated accuracy — significantly outperform static networks in promoting collective wisdom. Quality-weighted multi-platform aggregation implements this pattern: an adaptive organizational structure that addresses epistemic injustice (calibration overrides prestige), insouciance (financial stakes incentivize engagement), and fragmentation (price signals bypass hierarchical bottlenecks). A demographic limitation: the intersection of people who understand AI safety, prediction markets, and have resources to trade is small — likely dominated by Rationalist/EA communities with shared priors. See Appendix A.3 for quality weighting estimation methods.

⁴⁸As Section II.5 established and Appendix A.5 formalizes, AMTAIR's dashboard-only architecture degrades several randomization techniques from self-sufficient statistical mechanisms to mechanisms requiring organizational infrastructure. Embargo design is where this degradation is most consequential.

⁴⁹AMTAIR's argument structures — causal decompositions, conditional dependencies, node definitions — are the system's primary intellectual contribution and benefit from maximal openness. The probability outputs generated by combining these structures with forecast data are precisely what creates contamination risk. This represents partial decomposition of the constitutive tension — reducing probability-level contamination while leaving framing contamination intact — not its dissolution; the argument structures that must remain open are themselves a contamination channel. The separation requires different access policies, different governance structures, and different institutional commitments for each layer. Otherwise, it can be possible to reconstruct inputs, intermediate probabilities or outputs even from a limited set of released probabilities. Preventing this requires well thought out randomization algorithms.

structured awareness, not perfect separation, analogous to Chinese walls in financial regulation.⁵⁰ The deeper challenge is cultural: organizations transitioning from unquantified beliefs to explicit, verifiable probability claims face at-times uncomfortable transparency, and those most resistant are possibly those whose epistemic practices most need correction.

One intuitive corollary deserves explicit statement: where the organizational infrastructure for full hierarchical Bayesian integration is not yet available, naive integration with acknowledged limitations remains preferable to no integration at all.⁵¹

IV.3 Open Questions

Several challenges remain open, but each, on the organizational analysis developed here, is an organizational research question rather than a purely technical one.

Gaming and adversarial behavior. Thin AI safety prediction markets are vulnerable to manipulation, and the policy stakes can vastly exceed manipulation costs; partial defenses exist but none is complete.⁵²

Semantic mapping. Mapping AMTAIR nodes to forecasting questions requires human judgment⁵³ — another organizational process requiring governance — since a single node may map to multiple questions capturing different aspects and vice-versa.⁵⁴

Question selection. A value-of-information framework⁵⁵ should guide integration priorities, but prioritization itself requires organizational judgment about which decisions matter most.⁵⁶

Research agenda. Pilot implementations with historical data constitute immediate priorities; beyond mathematical refinements the substantive research program concerns institutional structures, governance design, and epistemic culture.⁵⁷

IV.4 Objections and Responses

Beyond open research questions, the approach faces three substantive objections.

⁵⁰The analogy carries empirical weight: evidence from financial regulation shows that Chinese walls, despite acknowledged permeability, measurably reduce insider trading. Clinical trial pre-registration provides a second existence proof: mandatory disclosure before data collection does not eliminate publication bias but creates accountability structures that significantly reduce it — imperfect organizational barriers producing measurable epistemic improvement. See Choo (2008) on structural impediment as organizational information failure.

⁵¹The hierarchical framework (A.1-A.5) requires α_i , γ_i , and organizational metadata that may not exist for all nodes, all platforms, or all time periods. Where these inputs are unavailable, the framework degrades gracefully: population-level corrections substitute for individual-level adjustments; qualitative contamination flags substitute for quantitative γ_i estimates; and at the limit, unweighted forecast aggregation — naive integration — still addresses parametric staleness even without contamination correction. The gradient from full hierarchical integration through partial correction to naive aggregation mirrors the addressability gradient in A.5: each step down sacrifices precision but retains the core benefit of external information. The appropriate response to missing infrastructure is not to abandon integration but to be transparent about what correction is and is not applied — reporting wider uncertainty bounds and flagging uncorrected contamination channels.

⁵²Track-record weighting, multi-platform aggregation, and anomaly detection provide partial defense. Proper scoring rules incentivize honest probability reporting; the deeper problem is that beliefs are genuinely contaminated, or that market structure permits strategic manipulation where a wealthy actor could move prices substantially with modest capital. See Appendix B.1 on mechanism design limitations.

⁵³This might be automatable with similar techniques to AMTAIR but doing so without introducing biases demands careful analysis.

⁵⁴The double-counting problem (conditional independence violations when contaminated forecasts are treated as independent evidence) is formally well-understood (Pearl, 2009) but its practical management depends on organizational tracking of information exposure. See Appendix A.1.

⁵⁵ $\text{VoI}(Q) = \mathbb{E}[\text{Value}(\text{Decision}|p^*)] - \mathbb{E}[\text{Value}(\text{Decision}|p)]$

⁵⁶Questions with large uncertainty reduction, decision-relevance, and well-specified resolution criteria deserve priority — but these criteria involve value judgments the statistical framework does not generate autonomously.

⁵⁷Specific organizational research questions include: How do institutional structures affect contamination rates? How does author refinement governance affect the quality-contamination trade-off? How does ensemble diversity interact with framing contamination? Embargo duration optimization and cascade detection metric calibration (Appendix A.4) are the technical complements.

A technical objection. One might object that there are privacy-preserving techniques that could in principle give AMTAIR the numerical secrecy that full platform control provides, dissolving the constitutive tension without requiring organizational infrastructure.⁵⁸ But as Sections I.2 and II.5 established, contamination operates through shared *framing*, not only through shared numbers. This objection addresses only the practical prong of the elimination argument. The principled prong — that framing contamination is inherently organizational regardless of control level — survives.

An organizational objection. A complementary worry runs in the opposite direction: organizational interventions might create an *illusion of control*, licensing continued integration under false confidence that contamination is managed.⁵⁹ This objection has force: it is precisely the recursive problem identified in Section III.3, where the managing organization cannot stand outside the epistemic dynamics it coordinates. The response is not to deny the risk but to note that informal integration will happen regardless (Section I.3), and unmanaged contamination without even cosmetic safeguards is worse than imperfect organizational management with known failure modes.

A scope objection. One might object that AMTAIR has not yet integrated prediction market data, making the elaborate organizational framework a solution to a problem that may never materialize at the scale envisioned.⁶⁰ Two responses apply. First, as Section I.3 illustrates, practitioners will be connecting prediction market outputs with Bayesian risk analyses informally; the organizational framework addresses integration that is happening *regardless* of AMTAIR's deployment status. Second, the design stage is precisely when organizational infrastructure should be planned; the most consequential failure modes (i.e. framing contamination and recursive capture) operate subtly and may be undetectable once pathways become entrenched.

V. Conclusion: Managing an Irreducible Tension

V.1 Summary

AMTAIR's contamination risk admits a powerful statistical response under idealized conditions, but the conditions require platform control that AMTAIR's open-loop architecture does not provide, and even full control leaves framing contamination intact. Under AMTAIR's actual constraints, practical statistical tools still help but each depends on organizational infrastructure the statistical framework requires but does not create. Epistemic vices, both individual and collective, explain why this infrastructure is unlikely to emerge spontaneously.⁶¹ The vice $\rightarrow$ vulnerability mapping (Section III.3) systematizes

⁵⁸Specific candidate techniques include differential privacy (adding calibrated noise to reported probabilities), cryptographic commitments (verifying forecast integrity without revealing values), and zero-knowledge proofs (demonstrating statistical properties without disclosing individual predictions). Each addresses the practical problem of numerical exposure. But AMTAIR's argument structures — causal decompositions and ontological choices — must be public for peer review and community contribution, and these structures are precisely the deeper contamination channel (the third contamination type identified in Section II.5). Privacy-preserving computation can obscure probabilities; it cannot obscure the framing through which those probabilities acquire meaning.

⁵⁹The concern is that embedding contamination management in institutional procedures produces cosmetic rather than substantive safeguards: embargo protocols exist on paper but nobody checks compliance; contamination audits become box-ticking exercises. The organizational framework developed here could make the situation *worse* than honest acknowledgment of irreducible contamination, by providing institutional cover for contamination that proceeds unchecked beneath official procedures.

⁶⁰This objection has particular force given that AMTAIR remains an academic thesis project without institutional adoption. The contamination dynamics analyzed in Sections II-III presuppose authority, broad uptake, and feedback loops that require a level of deployment AMTAIR has not achieved. If the system remains small, contamination risk may be negligible.

⁶¹Hubris overrides safeguards by treating model outputs as authoritative rather than provisional; insouciance neglects the metadata infrastructure on which contamination detection depends; injustice substitutes prestige for calibration in weighting forecaster contributions; and collective vices — fragmentation and complexity — prevent any single actor from seeing the full contamination loop.

these failures into design requirements. Section IV developed organizationally embedded responses: not solving the contamination problem but managing it through awareness of vulnerabilities and institutional infrastructure, themselves subject to the recursive dynamics Section III.3 identified.

V.2 Broader Implications

The constitutive tension is not unique to AMTAIR — analogous dynamics appear wherever institutional authority shapes the beliefs of those whose independent judgment the system needs.⁶²

It stands to reason that rapidly increasing AI capabilities will result in a proliferation of automated cognitive work, including scaffolding similar to the ones underlying AMTAIR. If such systems begin to increasingly build on their own outputs, entire information systems may risk confirmation loops, information cascades, contamination and other epistemic traps, similar to ones outlined here. The vice → vulnerability mapping's helpfulness in motivating institution-specific prescriptions from general analytical categories, suggests the method may transfer, though this remains an empirical question.

V.3 Critical Assessment

The framework has limitations which it may not fully escape.

Taxonomic incompleteness. The contamination taxonomy identifies three mechanisms but does not demonstrate exhaustiveness — Section III.1 acknowledges this and identifies temporal contamination as a further channel.⁶³

Generality as weakness. The constitutive tension's generality risks undermining the practical contribution — but generality of the *tension* does not entail generality of the *response*, which the vice → vulnerability mapping keeps institution-specific.⁶⁴

Absence of empirical validation. The framework is entirely theoretical — no contamination has been measured, no embargo tested — making pilot implementation the necessary next step.⁶⁵

Reflexive entanglement. The paper itself participates in the dynamics it analyzes: by publishing a contamination framework, it shapes how future implementers conceptualize the problem — a form of framing contamination operating through academic discourse. This is not a defeater, but the paper cannot stand fully outside the epistemic ecosystem it describes.⁶⁶

⁶²Examples include IPCC assessment cycles (where researchers on subsequent assessments internalize previous IPCC framings, creating citation cascades), WHO pandemic risk assessments (where model authority shapes the preparedness investments it evaluates), central bank economic forecasting (where published projections influence the market behavior they attempt to predict), and financial regulatory stress testing (where published scenarios become focal points for institutional planning). In each case, the recursive insight applies: the institution managing epistemic coordination is itself embedded in the dynamics it coordinates, and cannot claim to manage epistemic processes from a position outside them.

⁶³The three mechanisms were derived by asking what organizational infrastructure each statistical parameter requires — a productive heuristic, but one that is blind to contamination channels that do not correspond to named parameters in the hierarchical framework. A more systematic derivation might enumerate all causal pathways through which AMTAIR's outputs can influence the inputs it later ingests, then classify which are statistically addressable and which require organizational intervention. The present analysis follows the less rigorous path of identifying salient exemplars.

⁶⁴If every epistemic institution faces this tension, the organizational responses in Section IV may appear as generic institutional design principles dressed in AMTAIR-specific vocabulary. The response is that the vice→vulnerability mapping is specific to AMTAIR's architecture and constrains the space of adequate institutional responses, even though it does not uniquely determine them (Section IV develops one set of examples, not the only viable ones). Organizational epistemology contributes not generic principles but a *method* for producing institution-specific response constraints from system-specific vulnerability analysis. Whether this method transfers to other epistemic institutions is an empirical question this paper identifies but does not resolve.

⁶⁵The vice→vulnerability mapping in Section III.3 is analytically derived from applying Baird & Calvard (2019)'s and M. Meyer (2023)'s frameworks to AMTAIR's architecture, not empirically discovered through observing actual contamination dynamics. This is appropriate for a design-stage system — AMTAIR has not yet integrated prediction market data, so contamination has not yet occurred — but it means the framework's practical value is prospective rather than demonstrated. Pilot implementation with historical data (Section V.4) is the necessary next step; without it, the organizational prescriptions in Section IV remain plausible but untested.

⁶⁶If AMTAIR's developers adopt this paper's vice→vulnerability framework, they will see contamination through the categories developed here, potentially missing channels the framework does not name. The organizational epistemology tradition itself acknowledges this reflexivity: Choo (2008)'s framework for analyzing information environments is itself an

V.4 Looking Forward

Immediate next steps are pilot implementations with historical data, testing the methodology on resolved questions before deploying for live policy-relevant forecasts.⁶⁷ The longer-term agenda is organizational: building institutional infrastructure on which the statistical framework depends — an inverse problem for organizational design.⁶⁸

For AMTAIR specifically, the severity of most contamination dynamics identified in this paper scales with uptake: authority-driven anchoring presupposes authority, cascade formation requires a critical mass of exposed forecasters, and framing contamination propagates in proportion to the system's reach. At AMTAIR's current low baseline of adoption, this scaling creates a window for incremental institutional development; organizational infrastructure can be built as the system grows, rather than retrofitted after contamination pathways have become entrenched. This does not, however, license complacency: consequential failure modes may appear early and may scale quickly beyond when early intervention would be most effective.

Returning to the governance analyst and her brief. With organizationally embedded integration, she no longer faces a raw composite of AMTAIR's 35% *de novo* synthesis estimate versus the market's 72%. Instead, her dashboard helps her to decompose the signals⁶⁹: contamination auditing has identified which forecasters' reasoning draws on AMTAIR's decomposition framework, which reflects shared-literature correlation, and which carries genuinely new evidence (the insider evaluation data, the reproduction attempts, the expert biosecurity assessments that the market is uniquely positioned to aggregate). The effective sample size has been reduced to reflect the measured dependency; the ensemble's reported variance has been *widened* to account for residual contamination uncertainty rather than narrowed by the spurious convergence the unmanaged system would have produced. The prior 37-percentage-point gap between model and market is itself tracked as a learning signal, flagged for investigation rather than averaged away. She has *structured knowledge* of the factors driving divergence — which information channels are independent, which parameters are organizationally grounded, where uncertainty remains irreducible — even if definitive causal attribution exceeds what the framework can deliver under these circumstances. This structured uncertainty is organizationally produced, not statistically conjured, and it is categorically more useful than the unstructured confusion she started with.

The choice facing AMTAIR is not between integration and isolation; practitioners are likely to connect these epistemic resources informally regardless. The choice is between managed integration, with named vulnerabilities and organizational infrastructure designed to maintain the conditions that statistical tools require, and unmanaged integration, where contamination operates through channels no one tracks and no framework addresses. Organizational mitigations make the former categorically preferable.

What remains is the institutional work of building organizations that can sustain such commitments continuously, under the pressures of resource constraint, personnel turnover, and the very authority

information environment that shapes how analysts process organizational information. The appropriate response is not silence but self-awareness: this paper offers one analytically grounded decomposition of the contamination problem, not the uniquely correct one.

⁶⁷Backtesting (as discussed in Section III.1) can partially validate the approach where ground truth exists. But for the long-horizon questions most relevant to AI governance, ground truth may remain permanently unavailable, making the organizational dimensions of quality assessment all the more important.

⁶⁸Rather than asking "given this organization, what statistical methods work?" the question becomes "given the statistical methods we need, what organizational properties must we create?" The vice→vulnerability mapping provides the design specification; Section IV provides the implementation sketch. A complementary inverse problem operates at the epistemic level: systematic divergence between market prices and model outputs may function not merely as noise to be filtered but as a learning signal indicating where models need revision — transforming prediction markets from validation instruments into discovery instruments. Developing this bidirectional feedback direction requires the same organizational infrastructure discussed here: tracking divergence patterns rather than merely aggregating point estimates.

⁶⁹In such a scenario, analysts could use the BNs and model ensembles extracted with AMTAIR and update their inputs, conditional probabilities as well as adjusting the overall results.

dynamics this paper has analyzed.

References

- Almaatouq, A., Noriega-Campero, A., Alotaibi, A., Krafft, P. M., Moussaid, M., & Pentland, A. (2020). Adaptive social networks promote the wisdom of crowds. *Proceedings of the National Academy of Sciences*, 117(21), 11379–11386. <https://doi.org/10.1073/pnas.1917687117>
- Anderson, L. R., & Holt, C. A. (1997). Information Cascades in the Laboratory. *American Economic Review*, 87(5), 847–862. <https://doi.org/10.1257/aer.87.5.847>
- Baird, C., & Calvard, T. S. (2019). Epistemic Vices in Organizations: Knowledge, Truth, and Unethical Conduct. *Journal of Business Ethics*, 160(1), 263–276. <https://doi.org/10.1007/s10551-018-3897-z>
- Banerjee, A. V. (1992). A Simple Model of Herd Behavior. *The Quarterly Journal of Economics*, 107(3), 797–817. <https://doi.org/10.2307/2118364>
- Berea, A., Maxwell, D., & Twardy, C. (2012). Improving Forecasting Accuracy Using Bayesian Network Decomposition in Prediction Markets. *Proceedings of the AAAI Fall Symposium Series*. <https://www.aaai.org/ocs/index.php/FSS/FSS12/paper/view/5540>
- Berea, A., & Twardy, C. (2013). *Automated Trading in Prediction Markets*. 7812, 111–122. https://doi.org/10.1007/978-3-642-37210-0_13
- Bikhchandani, S., Hirshleifer, D., & Welch, I. (1992). A Theory of Fads, Fashion, Custom, and Cultural Change as Informational Cascades. *Journal of Political Economy*, 100(5), 992–1026. <https://doi.org/10.1086/261849>
- Brundage, M., Avin, S., Wang, J., Belfield, H., Krueger, G., Hadfield, G., Khlaaf, H., Yang, J., Toner, H., Fong, R., Maharaj, T., Koh, P. W., Hooker, S., Leung, J., & Dafoe, A. (2020). Toward Trustworthy AI Development: Mechanisms for Supporting Verifiable Claims. *arXiv Preprint*. <https://doi.org/10.48550/arXiv.2004.07213>
- Carlsmith, J. (2022). *Is power-seeking AI an existential risk?* <https://arxiv.org/abs/2206.13353>
- Celen, B., & Kariv, S. (2004). Distinguishing Informational Cascades from Herd Behavior in the Laboratory. *American Economic Review*, 94(3), 484–498. <https://doi.org/10.1257/0002828041464461>
- Choo, C. W. (2008). Information culture and information use: An exploratory study of three organizations. *Journal of the American Society for Information Science and Technology*. <https://doi.org/10.1002/asi.20797>
- de Bruin, B. (2013). Epistemic Virtues in Business. *Journal of Business Ethics*, 113(4), 583–595. <https://doi.org/10.1007/s10551-013-1677-3>
- Eyster, E., & Rabin, M. (2010). Naive Herding in Rich-Information Settings. *American Economic Journal: Microeconomics*, 2(4), 221–243. <https://doi.org/10.1257/mic.2.4.221>
- Frankfurt, H. G. (2005). *On Bullshit*. Princeton University Press. <https://doi.org/10.1515/9781400826537>
- Gelman, A. (2015, December 7). *Uncertainty laundering*. <https://statmodeling.stat.columbia.edu/2015/12/07/uncertainty-laundering/>
- Gneiting, T., & Raftery, A. E. (2007). Strictly Proper Scoring Rules, Prediction, and Estimation. *Journal of the American Statistical Association*, 102(477), 359–378. <https://doi.org/10.1198/016214506000001437>
- Karger, E., Bastani, H., Yueh-Han, C., Jacobs, Z., Halawi, D., Zhang, F., & Tetlock, P. E. (2024). ForecastBench: A Dynamic Benchmark of AI Forecasting Capabilities. *ICLR 2025*. <https://doi.org/10.48550/arXiv.2409.19839>
- Kokotajlo, D., Alexander, S., Larsen, T., Lifland, E., & Dean, R. (2025). *AI 2027: Scenario-based Forecasting for Transformative AI*. AI Futures Project. <https://ai-2027.com/>
- Kübler, D., & Weizsäcker, G. (2004). Limited Depth of Reasoning and Failure of Cascade Formation in

- the Laboratory. *The Review of Economic Studies*, 71(2), 425–441. <https://doi.org/10.1111/0034-6527.00290>
- Meyer, M. (2023). The Epistemic Vices of Corporations. *Synthese*, 201(5), 163. <https://doi.org/10.1007/s11229-023-04133-2>
- Meyer, V. J. (2025a). *Automating Transformative AI Risk Modeling* [Forthcoming, University of Bayreuth]. <https://vjmeyer.github.io/submission/>
- Meyer, V. J. (2025b). *From pre-paradigmatic chaos to constructive science: How AMTAIR could transform AI governance through evolutionary intervention* [Forthcoming]. University of Bayreuth. https://vjmeyer.github.io/submission/papers/2025-VJM_ParadigmFormation-AMTAIR_EvoRatPaper.html
- Meyer, V. J. (2025c). *Live models: Integrating prediction markets into AMTAIR’s bayesian risk networks: Managing contamination loops in collaborative forecasting systems* [Forthcoming]. University of Bayreuth. https://vjmeyer.github.io/submission/papers/2025-VJM_PredictionMarketIntegration-AMTAIR_EpiOrgsPaper.html
- Meyer, V. J. (2025d). *Practicing model pluralism: From expert intuitions to governance infrastructure* [Forthcoming]. University of Bayreuth. https://vjmeyer.github.io/submission/papers/2025-VJM_ModelPluralism-AMTAIR_PhilOfEconPaper.html
- Milgrom, P., & Stokey, N. (1982). Information, Trade and Common Knowledge. *Journal of Economic Theory*, 26(1), 17–27. [https://doi.org/10.1016/0022-0531\(82\)90046-1](https://doi.org/10.1016/0022-0531(82)90046-1)
- Pearl, J. (2009). *Causality: Models, reasoning and inference* (2nd ed.). Cambridge University Press.
- Powell, W. A., Hanson, R., Laskey, K. B., & Twardy, C. (2013). *Combinatorial Prediction Markets: An Experimental Study*. 8078. https://doi.org/10.1007/978-3-642-40381-1_22
- Ridder, J. de. (2022). Three models for collective intellectual virtues. In M. Alfano, C. Klein, & J. de Ridder (Eds.), *Social virtue epistemology* (pp. 335–356). Routledge.
- Schoenegger, P., Park, P. S., Karger, E., & Tetlock, P. E. (2024). Wisdom of the silicon crowd: LLM ensemble prediction capabilities rival human crowd accuracy. *Science Advances*. <https://doi.org/10.1126/sciadv.adp1528>
- Tetlock, P. E., & Gardner, D. (2015). *Superforecasting: The art and science of prediction* (First paperback edition). Broadway Books.
- Veit, W. (2019). Model Pluralism. *Philosophy of the Social Sciences*, 50(2), 91–114. <https://doi.org/10.1177/0048393119894897>
- Voigt, C. (2014). Argdown and the Stacked Masonry Layout: Two User Interfaces for Non-Expert Users. *Computational Models of Argument: Proceedings of COMMA 2014*, 483–484. <https://doi.org/10.3233/978-1-61499-436-7-483>
- Yudkowsky, E. (2015). *Rationality: From AI to zombies*. Machine Intelligence Research Institute.
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Appendices

Cross-Paper Integration Note: This paper (V. J. Meyer, 2025c) can be read independently. It is part of a three-paper thesis on AMTAIR. Companion papers address: (1) how model pluralism creates permanent trade-offs requiring formal coordination infrastructure (V. J. Meyer, 2025d); and (2) how formalization transforms pre-paradigmatic discourse toward greater coherence (V. J. Meyer, 2025b). All papers are available at <https://vjmeyer.github.io/submission/papers/> and at <https://www.amtair.org/publications/papers/>. Cross-references throughout indicate where content is developed further in companion papers.

Appendix A: Technical Specification

Extended Resources: Complete AMTAIR implementation details, thesis documentation, and interactive Colab notebooks demonstrating the methodology will be available upon thesis completion at the companion materials repository.

A.1 Formal Model

Let θ represent the parameter of interest (e.g., probability of transformative AI by 2030).⁷⁰ ⁷¹ Let M represent AMTAIR model outputs and $F = \{F_1, \dots, F_n\}$ represent forecasts from various platforms.⁷²

Standard updating (problematic):

$$P(\theta|F) \propto \prod_i P(F_i|\theta) \cdot P(\theta)$$

Quality-adjusted updating:

$$P(\theta|F, \alpha) \propto \prod_i P(F_i|\theta, \alpha_i) \cdot P(\theta)$$

where α_i encodes forecast quality parameters for source i . This addresses heterogeneous forecaster reliability but not contamination; the contamination correction model (γ_i , the exposure parameter) is

⁷⁰The complete hierarchical framework consists of two components presented separately: (1) quality-adjusted updating using α_i (this section and A.2-A.3), and (2) contamination correction using γ_i (A.5). A third parameter, the model-level extraction quality weights w_i (Section I.1), governs how AMTAIR's ensemble aggregates multiple extracted Bayesian networks; this is formalized in the companion paper (V. J. Meyer, 2025a) via Bayesian Model Averaging and is not developed here. The separate presentation mirrors the organizational decomposition — different institutions produce each parameter — though the correct statistical target is the joint model (see subsequent footnote).

⁷¹Because AMTAIR's extracted models are Bayesian networks with conditional probability structure $P(\theta_i | pa(\theta_i))$, prediction market data can in principle be integrated at multiple levels of the network: at input nodes (updating prior probabilities on root causes), at intermediate nodes (updating conditional relationships along causal pathways), at output nodes (comparing final posteriors with market aggregates), or at any combination thereof. Each integration point carries different contamination profiles and different organizational requirements. Input-level integration updates the least visible parameters and may produce the most informative corrections; output-level integration is most directly comparable to market prices but conflates multiple causal channels. The formal framework below uses θ generically to cover all such integration points; the choice of *where* in the network to inject market data is itself an organizational design decision with statistical consequences.

⁷²The notation $F = \{F_1, \dots, F_n\}$ assumes individual-level forecast observation — that AMTAIR can access each forecaster's submission separately. This holds for platforms providing individual-level data (e.g., Metaculus's public predictions, accessible via API with per-forecaster track records). It does not hold for platforms where AMTAIR observes only an aggregate market price $\bar{p}$ (e.g., Polymarket's AMM equilibrium). In the aggregate case, the contamination model (A.5) reduces to $\bar{p} = (1 - \bar{\gamma})\bar{p}^* + \bar{\gamma}p_A$, where $\bar{\gamma}$ is the exposure-weighted average contamination and $\bar{p}^*$ the uncontaminated consensus. Embargo-based estimation of $\bar{\gamma}$ survives (compare $\bar{p}$ during embargo vs. exposure periods), but per-forecaster quality weighting (α_i) and individual contamination estimation (γ_i) require individual-level data. Furthermore, market prices on AMM-based platforms are equilibrium outcomes of strategic trading, not simple averages of belief reports — the anchoring model's assumption that forecasters passively report probabilities is an additional idealization. These platform-level differences constitute another organizational design choice: AMTAIR's ability to implement the statistical framework depends partly on which platforms it partners with and how deeply it integrates with their APIs. See A.5 for the full treatment of how platform control level interacts with statistical technique availability.

developed in A.5, and the complete framework requires both.⁷³

A.2 Independence Conditions

Valid updating requires:

$$P(F_i|\theta, M) = P(F_i|\theta) \quad \forall i$$

The embargo mechanism aims to achieve this by ensuring forecasters cannot observe M before providing F_i .

A.3 Quality Parameter Estimation

Quality weights α_i are estimated from:

- Historical Brier scores
- Calibration curve analysis
- Market liquidity indicators⁷⁴

A.4 Cascade Detection Metrics

Contamination manifests as detectable statistical signatures in forecast data. The following metrics operationalize cascade detection:

Temporal correlation: Compare forecast distributions during embargo periods (AMTAIR outputs withheld) versus exposure periods. Statistically significant convergence toward AMTAIR estimates during exposure periods, absent during embargo periods, provides causal evidence of contamination. The embargo randomization enables difference-in-differences estimation.

Forecast clustering: Monitor the distribution of forecasts around AMTAIR's published probability. Under independence, forecaster distributions should reflect their diverse priors. Contamination produces anomalous clustering — excess probability mass near the published estimate relative to the pre-publication baseline.

Cross-platform divergence decay: Track how quickly forecast distributions across platforms (Metaculus, Manifold, Polymarket) converge after AMTAIR publication. Rapid cross-platform convergence exceeding what new information would explain signals contamination propagation through shared exposure rather than independent updating.

Framing contamination index: Measure whether forecasters' reasoning patterns — not just their probability estimates — shift toward AMTAIR's causal decompositions after exposure. This captures the deeper contamination channel identified in Section I.2: influence through shared framing rather than shared numbers. Operationalization requires natural language analysis of forecaster rationales, focusing on convergence in causal structure rather than surface-level language similarity.⁷⁵

These metrics inform the contamination parameter γ_i in the hierarchical framework (A.1-A.2) and provide the empirical foundation for the organizational contamination auditing proposed in Section IV.

⁷³This formalization treats quality (α_i) and contamination (γ_i , developed in A.5) as separate parameters in separate models. A fully unified formalization would model each forecast as $F_i \sim \mathcal{N}((1 - \gamma_i)\theta + \gamma_i p_A, \sigma^2(\alpha_i))$, jointly estimating reliability and contamination exposure. We present them separately because the paper's argument is that the organizational infrastructure required to *estimate* α_i (Section III.1: platform measurement systems) differs from that required to *estimate* γ_i (Section III.1: communication tracking) — the formalism mirrors the organizational decomposition. The joint model remains the correct statistical target; the separate presentation reflects the organizational reality that different institutions produce each parameter.

⁷⁴Estimation of contamination exposure (γ_i) is a separate problem addressed in A.5; this section concerns forecaster quality (α_i) only.

⁷⁵More precisely, the core measurement is dimensionality reduction in the hypothesis space $\mathcal{H}$: let $\dim(\mathcal{H})$ be the number of distinct causal decomposition frameworks invoked across the forecasting population. A drop in $\dim(\mathcal{H})$ signals framing contamination even when probability estimates remain diverse — ontological homogeneity masked by numerical disagreement. See A.5 dimension (iv) for full formalization.

A.5 Randomization, Platform Control, and the Contamination Signature

Section II.5 established that AMTAIR's dashboard-only architecture degrades the statistical toolkit, creating an organizational gap. This appendix develops the formal treatment: the full set of randomization techniques, their requirements under each control level, the multi-dimensional contamination signature, and the precise organizational infrastructure needed to bridge the gap.

Full platform control (idealized case). Suppose AMTAIR controlled the prediction market platforms — operating them directly, as DAGGRE operated within IARPA's competition platform. Five randomization techniques would then be available: (1) *temporal embargo*, withholding all AMTAIR outputs during randomly selected periods to enable difference-in-differences estimation of contamination effects; (2) *individual-level randomization*, randomly assigning which logged-in forecasters see AMTAIR outputs for each question, enabling per-forecaster γ_i estimation; (3) *staggered model publication*, releasing different BN models at different times to disentangle model-specific from general anchoring effects; (4) *output perturbation*, adding calibrated noise to published probability estimates (analogous to differential privacy) to bound the precision of any single anchoring signal; and (5) *partial node publication*, displaying structural decompositions while withholding probability estimates for randomly selected nodes. Together, these would make the contamination parameter γ_i experimentally estimable. Mean bias toward AMTAIR's published values becomes identifiable via (1) and (2). Variance distortion from correlated anchoring becomes measurable via (1). The effective sample size — $N_{\text{eff}} = (N - K) + K/(1 + (K - 1)\rho)$, where K forecasters share AMTAIR exposure at correlation ρ — becomes recoverable because K is known by construction.

Even under full control, however, one contamination channel persists. Techniques (1)–(4) randomize exposure to *probability estimates*. But as Section I.2 established, contamination also operates through shared *framing* — AMTAIR's causal decompositions structure how forecasters think about the problem. The structural decompositions must be public for peer review and community contribution (the design requirement established in Section II.1). Technique (5) partially addresses this, but withholding node probabilities while publishing the graph structure still exposes the ontological choices — which variables are included, how causal pathways are drawn, what counts as a relevant factor. Framing contamination is constitutive of AMTAIR's public infrastructure requirement and survives even maximal randomization capability.

Dashboard-only control (AMTAIR's actual architecture). AMTAIR does not control prediction market platforms. It publishes extracted BN models on its own dashboard and repository; it pulls forecast data from Metaculus, Manifold, and other platforms via their public APIs. The contamination channel is therefore indirect: AMTAIR dashboard $\rightarrow$ forecaster voluntarily visits $\rightarrow$ forecaster carries anchoring/framing to external platform $\rightarrow$ forecaster submits forecast $\rightarrow$ AMTAIR ingests via API. Each intermediary step attenuates and transforms the signal but also makes it harder to track.

Under this architecture, the five techniques degrade asymmetrically. Temporal embargo (1) remains fully feasible — AMTAIR controls when it publishes on its dashboard. Staggered publication (3), output perturbation (4), and partial node publication (5) also remain feasible for the same reason: AMTAIR controls its own dashboard content. Individual-level randomization (2), however, requires AMTAIR to build user authentication and selective display infrastructure on its own dashboard — feasible but substantially more complex than if AMTAIR controlled the platforms where forecasts are collected. More fundamentally, individual-level randomization captures only *direct dashboard visitors*. A single blog post, tweet, or workshop presentation summarizing AMTAIR's outputs contaminates forecasters who never visited the dashboard — a social-sharing channel that no dashboard-level randomization can control.

The critical quantity K (number of exposed forecasters) becomes unobservable without organizational tracking infrastructure. Under full platform control, K is known by construction. Under dashboard-only control, K depends on website analytics, social media monitoring, and community communication patterns — organizational facts that someone must design systems to collect, fund the infrastructure to maintain, and govern the institutional processes around. This is precisely the "insouciance" failure mode identified in Section III.2: the statistical framework needs K , but generating K requires sustained organizational investment in unglamorous metadata collection.

The multi-dimensional contamination signature. The contamination problem is not one-dimensional. At least five predictable distortions compose the full signature, and they are differentially addressable depending on platform control. Formalizing each reveals the structure of the problem and the specific organizational requirements for measurement.

- (i) *Mean bias toward AMTAIR's published values.* Let p_A denote AMTAIR's published probability and p_i^* forecaster i 's uncontaminated belief. Under anchoring, the submitted forecast becomes $\hat{p}_i = (1 - \gamma_i)p_i^* + \gamma_i p_A$, where $\gamma_i \in [0, 1]$ is the contamination exposure parameter (complementing the quality parameter α_i from A.1). The market mean shifts: $\mathbb{E}[\hat{p}] = \mathbb{E}[p^*] + \bar{\gamma}(p_A - \mathbb{E}[p^*])$, where $\bar{\gamma}$ is the exposure-weighted average contamination.⁷⁶ This is arguably the primary policy-relevant effect: if a governance body's risk assessment reduces to p_A with extra steps, the entire integration infrastructure provides false independence. The bias is systematic *per question* (always toward p_A), though its sign across questions depends on whether AMTAIR's prior exceeds or falls below the uncontaminated consensus. Addressable at full control via techniques (1)–(2), which enable estimation of $\bar{\gamma}$ through comparison of exposed and unexposed groups; partially addressable at dashboard-only control via (1), but $\bar{\gamma}$ estimation requires knowing K .
- (ii) *Variance underestimation through correlated anchoring and multi-round laundering.* Shared AMTAIR exposure induces pairwise correlation ρ among the K exposed forecasters' probability estimates. The effective sample size becomes:

$$N_{\text{eff}} = (N - K) + \frac{K}{1 + (K - 1)\rho}$$

For a market with $N = 100$ forecasters of whom $K = 40$ have visited AMTAIR's dashboard with average pairwise correlation $\rho = 0.5$: $N_{\text{eff}} = 60 + 40/20.5 \approx 62$, a 38% reduction from the nominal sample. Multi-round feedback amplifies this: if AMTAIR ingests the round- t market consensus and republishes, forecasters who anchor cumulatively (updating on top of previously contaminated beliefs rather than returning to fixed priors) produce recursive variance contraction among the exposed subpopulation:

$$\sigma_{t+1, \text{exposed}}^2 = (1 - \bar{\gamma})^2 \cdot \sigma_{t, \text{exposed}}^2 + \bar{\gamma}^2 \cdot \sigma_{\text{model}}^2 + 2\bar{\gamma}(1 - \bar{\gamma})\text{Cov}(\hat{p}^{(t)}, p_A^{(t)})$$

In the limiting case where AMTAIR republishes a point estimate ($\sigma_{\text{model}}^2 \rightarrow 0$) and the covariance term is small, variance among exposed forecasters decays geometrically at rate $(1 - \bar{\gamma})^2$ per round — after T rounds, $\sigma_{T, \text{exposed}}^2 \approx (1 - \bar{\gamma})^{2T} \sigma_0^2$.⁷⁷ Even modest contamination strength produces substantial

⁷⁶This clean decomposition assumes $\text{Cov}(\gamma_i, p_i^*) = 0$ — that contamination exposure is uncorrelated with uncontaminated beliefs. If forecasters who visit AMTAIR's dashboard systematically differ from those who do not (e.g., AI safety specialists with more extreme priors are also more likely to consult AMTAIR), the mean shift contains an additional selection-bias term: $\mathbb{E}[\hat{p}] = \mathbb{E}[p^*] + \bar{\gamma}(p_A - \mathbb{E}[p^*]) - \text{Cov}(\gamma_i, p_i^*)$. Individual-level randomization (technique 2) eliminates this selection bias by construction; population-level estimation from embargo data (technique 1 under dashboard-only control) does not.

⁷⁷For $\bar{\gamma} = 0.35$, the per-round contraction factor is $(1 - 0.35)^2 = 0.4225$; after 3 rounds, variance among exposed forecasters is approximately $0.4225^3 \approx 0.075$ of its original value — a 92% reduction. The *market-level* variance contracts more slowly

variance collapse over a few feedback cycles — uncertainty laundered through successive rounds, each cycle's tighter bounds becoming the next cycle's stronger anchor. Addressable at full control; partially addressable at dashboard-only control via embargo periods that interrupt the feedback cycle.

- (iii) *Variance overestimation from conflicting ensemble models.* AMTAIR's ensemble may publish multiple extractions with different probability estimates $(p_{A_1}, p_{A_2}, \dots)$. If some forecasters anchor on p_{A_1} and others on p_{A_2} , the market displays a mixture distribution: $\hat{p} \sim \sum_j w_j \cdot \mathcal{N}(p_{A_j}, \sigma_j^2)$, where w_j reflects the fraction of forecasters exposed to model j . The resulting variance $\text{Var}(\hat{p}) = \sum_j w_j \sigma_j^2 + \sum_j w_j (p_{A_j} - \bar{p}_A)^2$ includes a between-model component reflecting AMTAIR's internal disagreement rather than the forecasting community's genuine epistemic diversity. This effect operates in the opposite direction from (ii): while correlated anchoring on a single estimate *reduces* apparent variance, bimodal anchoring on conflicting estimates *inflates* it. Whether the net effect is under- or overestimation depends on the ensemble's publication format — a single averaged output triggers (ii); multiple displayed models trigger (iii). This is an organizational design choice about dashboard presentation, not a statistical parameter.
- (iv) *Dimensionality reduction in reasoning.* AMTAIR's causal decompositions constrain the hypothesis space forecasters explore. If AMTAIR decomposes $P(\text{catastrophe})$ into nodes representing capability overhang $\rightarrow$ alignment failure $\rightarrow$ deployment pressure, forecasters reason along these causal pathways rather than considering alternative decompositions (e.g., regulatory capture $\rightarrow$ competitive race $\rightarrow$ corner-cutting). Formally, let $\mathcal{H}$ denote the space of causal hypotheses a forecaster considers. Pre-exposure: $\dim(\mathcal{H}_i)$ varies across forecasters, reflecting diverse reasoning approaches. Post-exposure: $\mathcal{H}_i$ contracts toward $\mathcal{H}_A$ (AMTAIR's published decomposition), reducing the effective dimensionality of the collective hypothesis space even when point estimates diverge. This manifests as higher inter-forecaster correlation in *reasoning patterns* detectable via the framing contamination index (A.4) — forecasters may disagree on probabilities while agreeing on ontology, producing apparent diversity masking deep homogeneity. Partially detectable and mitigable — the framing contamination index (A.4) enables measurement, and randomized exposure to alternative decompositions can partially restore hypothesis-space diversity — but not eliminable at any control level. The structural decompositions must be public for peer review (Section II.1); publication constitutively shapes reasoning. This is why the constitutive tension survives even maximal statistical effort: what remains after detection, measurement, and partial correction is an organizational residual requiring cultural infrastructure rather than statistical technique.
- (v) *Asymmetric and temporally phase-dependent effects.* Anchoring strength is not constant: research on anchoring effects suggests that extreme anchors ($p_A \approx 0$ or $p_A \approx 1$) produce stronger contamination than moderate ones ($p_A \approx 0.5$), because the specificity and precision of an extreme estimate exerts more pull than a maximally uncertain one. Contamination therefore differentially affects the questions with the most extreme — and typically most policy-relevant — published estimates. The signature also evolves temporally across the feedback cycle: early rounds show strong mean bias (dimension i); middle rounds show variance collapse from multi-round laundering (dimension ii); late rounds may show mean reversion as new independent information enters, but with artificially tight confidence bounds from accumulated laundering. Formally, the contamination regime is char-

because the $N - K$ unexposed forecasters maintain their original variance: $\sigma_{t,\text{market}}^2 \approx \frac{N-K}{N} \sigma_0^2 + \frac{K}{N} \sigma_{t,\text{exposed}}^2$. With $K/N = 0.4$, the market variance after 3 rounds is approximately $0.6 \times \sigma_0^2 + 0.4 \times 0.075 \sigma_0^2 = 0.63 \sigma_0^2$ — a 37% reduction, less dramatic but still substantial. The K/N ratio thus determines the severity of market-level laundering: higher AMTAIR penetration produces faster collapse.

acterized by a time-varying vector $\mathbf{c}(t) = (\text{mean bias}(t), \text{variance distortion}(t), \text{dimensionality}(t))$, with different detection metrics (A.4) appropriate at different phases.

The table below summarizes addressability, using a spectrum — High, Partial, Minimal — rather than binary categories, because partial mitigation differs qualitatively from impossibility:

Dimension	Full Control	Dashboard-Only	Organizational Requirement
(i) Mean bias	High	Partial	Exposure tracking
(ii) Variance under	High	Partial	Embargo compliance
(iii) Variance over	High	High	Publication format governance
(iv) Dimensionality	Partial	Minimal	Cultural + structural ⁷⁸
(v) Asymmetric/temporal	High	Partial	Longitudinal monitoring

The pattern reveals a gradient, not a cliff. Dimensions (i)–(iii) are highly addressable under full control and partially addressable under dashboard-only control — statistical tools work well *if* organizational infrastructure supports them. Dimension (v) follows the same pattern with additional temporal complexity. Dimension (iv) is qualitatively different: detectable and partially mitigable even under full control, but with an irreducible organizational residual that no statistical technique can eliminate. Moving from full to dashboard-only control further degrades every dimension, shifting statistical mechanisms from self-sufficient to organizationally dependent. The gap — widening as one moves down the table from parameter correction to cognitive infrastructure — is filled by exactly the organizational commitments Section IV develops: exposure tracking (against insouciance), embargo enforcement (against hubris), publication governance (against complexity), longitudinal contamination auditing (against fragmentation), and for the deepest contamination channel, a culture of actively soliciting alternative ontologies. The statistical tools remain necessary; the organizational framework makes them possible; for framing contamination, the organizational framework is most of what is possible.

A further degradation: aggregate versus individual observation. The analysis above assumes AMTAIR observes individual forecasts F_i . In practice, observation level varies across platforms. Metaculus provides individual predictions with per-forecaster track records (closest to the formalism's assumptions). Manifold Markets provides individual bet histories but not comparable quality metrics. Polymarket provides primarily an aggregate market price — an AMM equilibrium reflecting strategic trading behavior, not a simple average of probability beliefs. When AMTAIR observes only an aggregate price $\bar{p}$, the contamination model collapses to its population-level form:

⁷⁸Under full control, randomized exposure to alternative decompositions, pre-exposure elicitation of forecasters' own causal models, and diversity indices on the hypothesis space $\mathcal{H}$ can detect and partially mitigate dimensionality reduction. The framing contamination index (A.4) provides the measurement basis. Under dashboard-only control, these techniques degrade further — AMTAIR cannot control which decompositions forecasters encounter through social channels. The residual is irreducible: the structural decompositions must be public for peer review (Section II.1), and publication constitutively shapes reasoning. What survives maximal statistical effort is organizational — a culture of actively soliciting alternative decompositions and rewarding ontological diversity.

$$\bar{p} = (1 - \bar{\gamma})\bar{p}^* + \bar{\gamma}p_A$$

where $\bar{\gamma}$ is the exposure-weighted average contamination and $\bar{p}^*$ the uncontaminated consensus price. Some techniques survive this aggregation: temporal embargo (technique 1) still enables difference-in-differences estimation of $\bar{\gamma}$ by comparing $\bar{p}$ during embargo versus exposure periods. But per-forecaster quality weighting (α_i), individual contamination estimation (γ_i), and the decomposition of N_{eff} into K exposed and $N - K$ independent forecasters all require individual-level data that aggregate prices do not provide. The cross-platform divergence metric (A.4) also requires platform-level aggregates at minimum, though not individual forecasts. This introduces a new organizational design choice: AMTAIR's statistical capability depends partly on which platforms it integrates with and how deeply. Prioritizing platforms that expose individual-level data (Metaculus's API) over those providing only aggregate prices (Polymarket) is itself an organizational decision with epistemic consequences — a further instance of statistical requirements generating organizational dependencies.⁷⁹

A.6 From Formalism to Practice: An Analyst's Guide

The formal apparatus developed in A.1–A.5 specifies what valid integration *requires*. This section traces what it *looks like* from the perspective of an analyst using AMTAIR, and where the gap between formal prescription and practical reality opens. The vignette's governance analyst (Section I.1) serves as the running example: she has AMTAIR reading 35% on bio-uplift, markets reading 72%, and days to brief.

Step 1: What the analyst wants. The analyst's target is $P(\theta \mid M, F)$ — the probability of genuine bio-uplift given both AMTAIR's model output M (the 35% with its five-hypothesis decomposition) and forecast data F from prediction markets (the 72% and surrounding distribution). The standard Bayesian update (A.1) says: multiply each forecast's likelihood by the prior and normalize. If she could do this naively — treat the 72% as independent evidence and combine it with the 35% — she would get something around 50-55%, and that would be her brief.

Step 2: Why she cannot do this. The independence condition (A.2) states: $P(F_i \mid \theta, M) = P(F_i \mid \theta)$. In words: each forecast must carry information about reality, not about AMTAIR's output. But AMTAIR published its dashboard a week ago. Biosecurity Twitter discussed the five-hypothesis decomposition. A Metaculus forecaster wrote a detailed rationale citing AMTAIR's scaffolding-contingent capability analysis. The 72% already contains reflections of the 35%. Naively combining them double-counts AMTAIR's signal. The resulting estimate would look like genuine convergence between model and market — but some of that convergence is an echo.

How this manifests concretely: the analyst's brief would report higher confidence than the evidence warrants. Policymakers would see "both our models and the markets agree," when in reality the markets partly agree *because they saw the models*. A decision to impose emergency restrictions that should carry a wide uncertainty band instead arrives with false precision. The contamination is invisible in the headline numbers; it lives in the covariance structure.

Step 3: What the hierarchical framework prescribes. The quality-adjusted update (A.1) replaces the naive product with quality-weighted integration:

⁷⁹An additional idealization: even on platforms providing individual forecasts, the anchoring model ($\hat{p}_i = (1 - \gamma_i)\bar{p}^* + \gamma_i p_A$) treats forecasters as passive probability reporters. On trading platforms, submitted "forecasts" are bet positions shaped by risk aversion, portfolio effects, and strategic timing — not pure belief reports. The gap between strategic trading behavior and the probability-reporting assumption is a further source of model misspecification, distinct from the contamination problem but compounding it. Cross-platform comparison (Metaculus's belief reports vs. Polymarket's trading prices) can partially disentangle these effects, provided AMTAIR has the organizational infrastructure to maintain integrated multi-platform data pipelines.

$$P(\theta | F, \alpha) \propto \prod_i P(F_i | \theta, \alpha_i) \cdot P(\theta)$$

Each forecast F_i is weighted by α_i , the forecaster's reliability. A superforecaster with a decade of calibrated predictions on biosecurity questions gets high weight; a newcomer with no track record gets low weight. The contamination model (A.5) then adjusts each forecast for exposure:

$$\hat{p}_i = (1 - \gamma_i)p_i^* + \gamma_i p_A$$

The submitted forecast $\hat{p}_i$ is a mixture of the forecaster's genuine belief p_i^* and AMTAIR's published value $p_A = 0.35$. If $\gamma_i = 0.4$ for a forecaster who submitted 0.58, their uncontaminated belief was approximately $p_i^* = (0.58 - 0.4 \times 0.35)/0.6 \approx 0.73$ — substantially higher than their submission suggested, and much further from AMTAIR's estimate.

What this looks like under idealized conditions (full platform control). Suppose AMTAIR controlled Metaculus and could run experiments. It would:

- (a) *Randomly embargo half the forecasters* (technique 1): 50 forecasters see AMTAIR's dashboard, 50 do not. Compare distributions. The exposed group clusters around 0.63; the unexposed group centers on 0.78. Difference-in-differences gives $\hat{\gamma} \approx 0.35$ — the average contamination strength.⁸⁰ The analyst now knows: roughly 35% of each exposed forecaster's submitted probability reflects AMTAIR's anchor rather than independent judgment.
- (b) *Estimate individual γ_i* (technique 2): By tracking which logged-in users visited which dashboard pages, AMTAIR labels each forecaster's exposure level. A forecaster who spent 20 minutes examining the five-hypothesis decomposition has higher γ_i than one who glanced at the headline number. The hierarchical model downweights accordingly.
- (c) *Recover the effective sample size*: With $N = 100$ forecasters, $K = 40$ exposed at average pairwise correlation $\rho = 0.5$:

$$N_{\text{eff}} = 60 + \frac{40}{1 + 39 \times 0.5} = 60 + 2.0 = 62$$

The market of 100 forecasters has the information content of 62 independent ones. The analyst's confidence intervals⁸¹ widen by $\sqrt{100/62} \approx 1.27$ — a 27% widening that turns a seemingly precise estimate into a more honest one. Her brief now says: "Markets indicate 72%, but correcting for AMTAIR contamination, the independent market signal is approximately 78% with substantially wider uncertainty than the raw data suggests."

- (d) *Detect variance collapse across rounds*: AMTAIR published version 1 of its bio-uplift analysis three months ago. Market variance has contracted each month. The multi-round laundering model (A.5) predicts recursive contraction: among the $K = 40$ exposed forecasters, variance decays at

⁸⁰Derivation: if the uncontaminated mean is $\bar{p}^* = 0.78$ (the unexposed group) and AMTAIR published $p_A = 0.35$, then under the anchoring model the exposed group's mean is $(1 - \bar{\gamma}) \times 0.78 + \bar{\gamma} \times 0.35 = 0.78 - 0.43\bar{\gamma}$. Setting this equal to the observed 0.63 gives $\bar{\gamma} = 0.15/0.43 \approx 0.35$.

⁸¹This paper uses "confidence interval" throughout in the conventional forecasting sense — the interval within which the true value is expected to lie with stated probability. Strictly, AMTAIR's Bayesian network outputs are posterior distributions, making their intervals "credible intervals" in the Bayesian sense, while prediction market aggregates are distributional summaries of crowd forecasts for which neither frequentist "confidence interval" nor Bayesian "credible interval" is precisely correct. We follow forecasting convention rather than introducing terminological distinctions that would complicate exposition without altering any substantive claim.

approximately $(1 - \bar{\gamma})^2 \approx 0.42$ per round, so after $T = 3$ monthly cycles, their variance has collapsed to roughly $0.42^3 \approx 0.075$ of its original value. The market-level variance is buffered by the 60 unexposed forecasters but still contracts substantially: approximately $0.6 + 0.4 \times 0.075 = 0.63$ of the true variance — a 37% reduction.⁸² The analyst would note: "The tight clustering of forecasts around 72% partly reflects successive rounds of anchoring, not independent convergence."

- (e) *Test for framing contamination* (dimension iv): Compare the causal hypotheses forecasters invoke in their rationales. Pre-exposure, the community considered seven distinct decomposition frameworks. Post-exposure, 85% of rationales use AMTAIR's five-hypothesis structure. Hypothesis-space dimensionality dropped from $\dim(\mathcal{H}) = 7$ to $\dim(\mathcal{H}) \approx 2$, measured as the number of frameworks accounting for $> 5\%$ of rationales (AMTAIR's framework at 85% plus one alternative at 15%; the remaining five frameworks have effectively disappeared from the reasoning population). Probability estimates diverge (35% to 80%), but ontological homogeneity masks the loss of reasoning diversity. The framing contamination index (A.4) captures this: high inter-forecaster correlation in reasoning patterns despite apparent disagreement in point estimates.

What this looks like under AMTAIR's actual architecture (dashboard-only control).

Each idealized capability degrades:

- (a) *Embargo*: AMTAIR can withhold its own dashboard publications. It *cannot* prevent a blogger from summarizing its last update, a workshop presenter from showing its decomposition, or a Twitter thread from propagating its framing. Embargo produces a clean "unexposed" comparison group only for *direct dashboard visitors*. Indirect exposure through social channels is untrackable at dashboard level. The analyst knows some forecasters were unexposed during the embargo window, but not how many — K becomes a range, not a number.

Practically: AMTAIR publishes on a Tuesday. A well-known AI safety researcher tweets a screenshot of the five-hypothesis decomposition within hours. By Wednesday, forecasters who never visited AMTAIR's dashboard are reasoning within its categories. The embargo comparison is contaminated by social propagation. The difference-in-differences estimate of $\bar{\gamma}$ captures only the direct-exposure component — indirect framing contamination goes unmeasured.

- (b) *Individual γ_i* : Without platform-level user tracking, AMTAIR cannot label which forecasters visited its dashboard. It could build authentication and analytics on its own site (feasible but non-trivial engineering), but this captures only direct visitors. The analyst must work with population-level $\bar{\gamma}$ estimated from embargo periods, not individual γ_i . Quality weighting falls back to whatever track-record data platforms provide through their public APIs — Metaculus Brier scores, Manifold calibration data — rather than integrated contamination-adjusted quality scores.

Practically: The analyst sees 100 forecasters on Metaculus. She can pull their Brier scores (platform provides this). She cannot tell which ones visited AMTAIR's dashboard. She uses the population-level $\bar{\gamma}$ from the last embargo comparison (say, 0.25 — lower than the idealized estimate because it misses indirect exposure) and applies it uniformly. The correction is better than nothing but coarser than individual-level adjustment.

⁸²This assumes cumulative anchoring (exposed forecasters update on top of previously contaminated beliefs) and that AMTAIR republishes a relatively precise estimate each round. If forecasters return to fixed prior beliefs each round rather than anchoring cumulatively, the per-round variance among exposed forecasters is constant at $(1 - \bar{\gamma})^2 \sigma_0^2$ and multi-round collapse does not occur. The empirical question — whether anchoring is cumulative or reset each round — has significant practical consequences and is itself resolvable only through the longitudinal monitoring infrastructure discussed in Section IV.

- (c) *Effective sample size*: Without knowing K , the N_{eff} formula becomes uncomputable. The analyst can estimate a range: if K is between 20 and 60 (based on dashboard analytics showing 30 unique visitors plus an unknown number of indirect exposures), N_{eff} is between 45 and 82. Her brief reports this as an uncertainty range rather than a point estimate — honest but substantially less useful.
- (d) *Variance collapse*: Embargo periods interrupt the feedback cycle, which is the primary practical defense. But the analyst cannot directly measure the per-round contraction rate because both $\bar{\gamma}$ and K carry uncertainty from (a). She can observe that variance has contracted over months and attribute some of that to laundering, but cannot decompose how much is genuine evidence accumulation versus how much is echo. The qualitative diagnosis (some laundering is occurring) is available; the quantitative diagnosis (exactly how much) is not.

Practically: Three months of shrinking confidence intervals on Metaculus. The analyst's supervisors see this as the market "converging on an answer." She suspects laundering but cannot prove how much. Her brief hedges: "Market confidence has increased, but some of this tightening may reflect successive rounds of anchoring on AMTAIR's estimates rather than independent evidence accumulation."

- (e) *Framing contamination*: The analyst can run NLP analysis on Metaculus rationales (public text data) and measure whether AMTAIR's five-hypothesis structure dominates post-publication. This detection capability survives dashboard-only control — it requires access to forecaster rationales, not platform control. But mitigation is limited. AMTAIR could solicit alternative decompositions on its own dashboard (organizational intervention), incentivize novelty in community contributions, or explicitly label its decomposition as one among many. None of these require platform control. All of them require sustained organizational commitment.

Practically: The analyst examines 50 Metaculus rationales. Before AMTAIR's publication, forecasters used diverse frameworks: some reasoned about model architecture, some about training data, some about evaluation methodology, some about geopolitical context. After publication, 40 of 50 rationales reference AMTAIR's five hypotheses by name. The community has adopted AMTAIR's ontology. Probability estimates still vary (35% to 80%), but the *reasoning space* has collapsed. Her brief should note this: "Market participants display diverse probability estimates but homogeneous causal frameworks, largely derived from AMTAIR's published decomposition. The diversity of point estimates may overstate the genuine independence of the underlying reasoning."

Step 4: One approach an analyst might take.

The following illustrates one plausible workflow under dashboard-only control, incomplete γ_i estimation, and unquantifiable framing contamination. It is not the only approach — alternative statistical methods (see below) could substitute for or complement several steps — but it makes the organizational dependencies concrete:

1. *Start with AMTAIR's decomposition* (35% bio-uplift), noting the five-hypothesis structure and its confidence intervals.
2. *Pull Metaculus individual predictions via API*. Compute the raw market distribution. Note the clustering pattern — if forecasts cluster bimodally (around AMTAIR's 35% and around 75%), this suggests two populations: AMTAIR-influenced and independent.
3. *Apply population-level contamination correction*. Using the $\bar{\gamma}$ estimate from the last embargo period

(say, 0.25), shift the market mean: $\bar{p}^* \approx (\bar{p} - \bar{\gamma} \cdot p_A) / (1 - \bar{\gamma}) = (0.72 - 0.25 \times 0.35) / 0.75 \approx 0.84$.⁸³ The corrected independent market signal is approximately 84% — substantially higher than the raw 72%, and much further from AMTAIR's 35%.

4. *Widen confidence intervals.* Without knowing N_{eff} precisely, use the plausible range (45-82) to bound the uncertainty. Report the corrected estimate with wider intervals than the raw market data suggests.
5. *Flag framing contamination qualitatively.* Note in the brief that market reasoning patterns converge on AMTAIR's decomposition, reducing the effective independence of forecasts even after numerical correction. Recommend that policymakers seek assessment from analysts who have NOT seen AMTAIR's framework — a concrete organizational recommendation the formal apparatus motivates but cannot implement.
6. *Report the gap.* The brief should state what the analyst *can* and *cannot* correct for. Mean bias: partially corrected via population-level $\bar{\gamma}$. Variance distortion: flagged qualitatively, bounds estimated. Framing contamination: detected but not corrected. The honest uncertainty band around the combined estimate is wider than either AMTAIR or the market alone would suggest — and that widening is the framework's most important practical output.

The organizational dependencies, made concrete. The table below maps each step the analyst takes to the organizational infrastructure it requires — infrastructure that the statistical framework prescribes but does not produce.

Analyst Step	Statistical Requirement	Organizational Dependency	Who Provides It
Quality weighting	α_i per forecaster	Track-record databases	Metaculus, Manifold (platform orgs)
Contamination correction	$\bar{\gamma}$ from embargo	Embargo enforcement + dashboard analytics	AMTAIR operators
Individual adjustment	γ_i per forecaster	User authentication + exposure logging	AMTAIR + platform partnership
Sample size correction	K, ρ estimates	Exposure tracking across channels	Cross-organizational audit
Variance collapse detection	Multi-round σ_t^2 series	Longitudinal data pipeline maintenance	AMTAIR data team
Framing contamination detection	NLP on rationale texts	Rationale archival + analysis infrastructure	Platform APIs + analysis team
Framing contamination mitigation	Alternative decomposition elicitation	Community engagement + incentive design	AMTAIR governance
Cross-platform synthesis	Integrated multi-platform data	API maintenance + format harmonization	AMTAIR engineering

⁸³This correction assumes $\text{Cov}(\gamma_i, p_i^*) = 0$ — that contamination exposure is uncorrelated with uncontaminated beliefs. If AI safety specialists with more extreme priors are also more likely to consult AMTAIR (plausible selection bias), the true uncontaminated mean is higher than 0.84. See A.5(i) footnote for the exact correction term.

Every row requires someone to build, fund, maintain, and govern the infrastructure. None of it exists automatically. The analyst's ability to produce a sound brief depends on an organizational supply chain stretching across multiple institutions — and the weakest link in that chain determines the quality of the statistical output.

Alternative statistical approaches. The workflow above uses the hierarchical Bayesian framework (A.1) with embargo-based contamination estimation — but this is not the only statistical strategy. Several alternatives could substitute for or complement individual steps:

- *Finite mixture models* can decompose the forecast distribution into contaminated and uncontaminated subpopulations without requiring individual exposure labels — partially bypassing the K -observability problem. But they require the analyst to specify the number of mixture components (an assumption) and enough distributional separation to identify the subpopulations (an empirical condition that may not hold when contamination is subtle).
- *Instrumental variable estimation* could identify contamination effects if an instrument exists that predicts forecaster behavior but is uncorrelated with the contamination channel — for example, a platform's interface change that altered dashboard visibility without affecting information content. Finding valid instruments is an organizational and empirical challenge, not a statistical one.
- *Propensity score matching* could pair contaminated and uncontaminated forecasters on observable characteristics (expertise domain, historical calibration, platform tenure) to estimate the contamination effect. This requires individual-level data on both treatment (exposure) and covariates — more organizationally demanding than the population-level approach above, not less.
- *Copula models* can represent the dependence structure among forecasts directly, without requiring the independence assumption. But specifying the copula requires information about *why* forecasts are dependent — which brings the analysis back to organizational facts about exposure channels.
- *Causal discovery algorithms* (PC, FCI) applied to forecast data could identify contamination pathways from observational data alone, without experimental randomization. These methods require large samples and the causal Markov assumption — and their outputs identify *that* a causal pathway exists, not *what organizational process produces it*.

The pattern across alternatives: each substitutes a different statistical assumption for the organizational inputs the hierarchical framework requires, but none eliminates organizational dependency entirely. Mixture models trade exposure labels for distributional assumptions. IV estimation trades randomization for instrument validity (an organizational-empirical judgment). Propensity scores trade population-level estimation for richer individual-level data. Copulas trade independence for dependence-structure knowledge. Causal discovery trades experimental control for sample size and Markov assumptions. The organizational gap is not an artifact of the hierarchical Bayesian approach; it is structural to the contamination problem. Any statistical method that addresses contamination must, at some point, interface with organizational facts about how information flows — because contamination *is* an organizational phenomenon with statistical consequences, not a statistical phenomenon with organizational decorations.

This is the paper's core claim made operational: the math works; what makes it *work in practice* is organizational.

Appendix B: Extended Literature Review

See Also: The AMTAIR thesis ([V. J. Meyer, 2025a](#)) provides comprehensive literature review across information cascades, organizational epistemology, and AI forecasting. Specific course

readings from "Epistemology for Organizations" (UBT, 2022) informed the organizational analysis throughout this paper.

B.1 Cascade Theory

The information cascade literature extends beyond Bikhchandani et al. (1992). Key developments relevant to AMTAIR's contamination problem include:

- Banerjee (1992): Independent discovery of cascade dynamics through a distinct model (agents choose sequentially from a discrete action set). Where Bikhchandani et al. (1992) emphasize cascade *fragility* — cascades break when a sufficiently informative signal arrives — Banerjee's model produces *robust* herding when the action space is coarser than the signal space, a distinction relevant to AMTAIR: forecasters submitting point probabilities (fine action space) can in principle break cascades, but forecasters adopting AMTAIR's causal decomposition (coarse framing choice) cannot
- Anderson & Holt (1997): Experimental verification of cascade formation in laboratory settings, confirming that cascades form even when subjects are Bayesian and information is unambiguous — a finding that undermines the hope that "rational markets self-correct"
- Kübler & Weizsäcker (2004): Heterogeneous information quality effects on cascade dynamics; agents with higher-quality private signals are more resistant to cascades but can still be swept into them, directly relevant to AMTAIR's quality-weighting approach
- Çelen & Kariv (2004): Sequential social learning with endogenous timing; agents who can choose *when* to act partially mitigate cascades by waiting for more information — relevant because prediction market forecasters can choose when to update, unlike the forced-sequential models
- Eyster & Rabin (2010): "Cursed equilibrium" in which agents naively equate others' actions with their private information, generating excess herding relative to Bayesian predictions; empirically more descriptive of actual market behavior than fully rational cascade models
- Almaatouq et al. (2020): Adaptive social networks and collective accuracy — demonstrating that network structure determines whether social information helps or harms crowd wisdom, directly supporting quality weighting mechanisms as organizational anti-cascade architecture

B.2 Organizational Epistemology

Course-relevant readings informing this analysis:

- Baird & Calvard (2019, pp. 264–272): Epistemic vices in organizational contexts; how organizational structure amplifies individual epistemic characteristics
- Choo (2008, pp. 32–42): Information use environments; organizations as information processors with inherent feedback dynamics
- de Bruin (2013): Epistemic virtues in business contexts
- M. Meyer (2023, Sect. 4–5, Table 2): Empirical identification of six corporate epistemic vices from 1.3 million employee reviews

Vice-Countermeasure Mapping. Sections II.3 and III.2–III.3 develop the integrated argument: PM-BN integration constitutes epistemic architecture against organizational vices (II.3), while those same vices explain specific contamination failure modes and why statistical safeguards break down (III.2–III.3). The complete mapping between the vices identified by Baird & Calvard (2019) and M. Meyer (2023) and the specific mechanisms proposed in this paper is as follows: *Epistemic malevolence* (willful opposition to truth) is countered by financial incentives rewarding accuracy — organizations tolerating motivated

reasoning gain an external check when market prices expose systematically wrong beliefs. *Epistemic insouciance* (indifference to accuracy; cf. Frankfurt (2005) on "bullshit") is corrected by forced numerical commitments and coherence requirements within the hierarchical Bayesian framework. *Epistemic hubris* (overconfidence) is exposed through calibration tracking and quality weighting — forecasters with poor track records receive diminished influence regardless of status. *Epistemic injustice* (status-based discounting of knowers) is structurally bypassed by track-record-based weighting that inverts traditional organizational hierarchies, giving a well-calibrated junior analyst more influence than a poorly calibrated senior executive. From M. Meyer (2023)'s empirically-derived taxonomy: *fragmentation* is addressed through cross-platform aggregation synthesizing distributed knowledge across organizational boundaries; *complexity* is countered by transparent BN methodology making causal assumptions auditable.

B.3 AI Forecasting Benchmarks

Recent developments in LLM forecasting:

- Karger et al. (2024, Sects. 3-5): ForecastBench systematic benchmarking methodology
- Schoenegger et al. (2024): Ensemble methods for AI forecasting

Appendix C: Example Extraction — Carlsmith (2022)

Cross-Paper Note: The companion paper (V. J. Meyer, 2025d) presents the full 23-node ArgDown extraction with complete BayesDown probability tables. This appendix provides a condensed version sufficient to make the contamination analysis concrete. For format specifications, see that paper's Appendix A; for the full extraction, source code, and interactive demonstrations, see V. J. Meyer (2025a) and the public repository.

AMTAIR's two-stage pipeline — ArgDown extraction followed by BayesDown enrichment — is illustrated here through the extraction of Carlsmith (2022)'s "Is Power-Seeking AI an Existential Risk?" Note: this extraction (six conjunctive conditions for AI existential risk) differs from the bio-uplift decomposition in Section I.1's vignette (five hypothesis classes: de novo, recombinative, elicitation artifact, memorization, scaffolding). AMTAIR's extraction approach yields different decompositions for different research questions; both are valid formalizations of their respective source arguments. This example makes the contamination discussion concrete: the specific probability estimates shown below are what forecasters would observe on AMTAIR's dashboard and potentially anchor on.

Condensed ArgDown Structure. The extraction identifies a 23-node causal hierarchy. The main chain decomposes Carlsmith's argument into six conjunctive conditions:

1. Timely development of advanced AI systems — $P(\text{TRUE}) = 0.80$
2. Some systems exhibit power-seeking behavior — $P(\text{power-seeking} \mid \text{development}) = 0.60$
3. Sufficient capability for existential-scale impact — $P(\text{capability} \mid \text{power-seeking, development}) = 0.50$
4. Power-seeking is misaligned with human values — $P(\text{misalignment} \mid \dots) = 0.70$
5. Deployment despite misalignment risk — $P(\text{deployment} \mid \dots) = 0.60$
6. Misaligned deployment leads to catastrophe — $P(\text{catastrophe} \mid \dots) = 0.70$

Overall $P(\text{existential catastrophe}) = \text{product of conditionals} \approx 0.05\text{--}0.10$ (5–10%).⁸⁴

BayesDown Probability Table (Excerpt). The hub node illustrates conditional probability table complexity — its three parents yield eight conditional entries:

```
[Misaligned_Power_Seeking]: Deployed AI systems seeking power
in unintended and high-impact ways.
{"instantiations": ["TRUE", "FALSE"],
 "priors": {"p(TRUE)": "0.338"},
 "posteriors": {
   "p(TRUE|aps=T, align_hard=T, deploy=DEPLOY)": "0.90",
   "p(TRUE|aps=T, align_hard=T, deploy=WITHHOLD)": "0.10",
   "p(TRUE|aps=T, align_hard=F, deploy=DEPLOY)": "0.25",
   "p(TRUE|aps=T, align_hard=F, deploy=WITHHOLD)": "0.05",
   "p(TRUE|aps=F, ...)": "0.0"}}
```

When APS systems do not exist ($\text{aps}=\text{F}$), $P(\text{misaligned power seeking}) = 0.0$ regardless of other factors — advanced capabilities are a necessary precondition. When all three risk factors align ($\text{aps}=\text{T}$, $\text{align_hard}=\text{T}$, $\text{deploy}=\text{DEPLOY}$), $P = 0.90$.

⁸⁴The straightforward product of the point estimates yields $0.80 \times 0.60 \times 0.50 \times 0.70 \times 0.60 \times 0.70 \approx 0.035$ (3.5%). The higher 5–10% range reflects Carlsmith (2022)'s own estimate, which accounts for uncertainty distributions around each parameter and conditional dependencies among the six conditions that make the simple product a lower bound. AMTAIR extracts the range rather than the point product.

Contamination Relevance. These structured outputs are precisely what creates the contamination risk analyzed in this paper. A forecaster viewing "P(misaligned power seeking) = 0.338" on AMTAIR's dashboard receives a specific anchor. More subtly, AMTAIR's *chosen decomposition* — representing Carlsmith's argument as a conjunction of APS systems, alignment difficulty, and deployment decisions — constitutes framing contamination (Section I.2): forecasters may adopt this extracted causal structure even without consulting specific probabilities, and the structure itself may differ from how Carlsmith organized his original argument. The cascade detection metrics in Appendix A.4 are designed to detect both channels.

Appendix D: Reference Tables

D.1 List of Abbreviations

Abbreviation	Full Term
ACE	Aggregative Contingent Estimation
AGI	Artificial General Intelligence
AI	Artificial Intelligence
AMTAIR	Automated Modeling of Transformative AI Risks
APS	Advanced, Planning, Strategically-aware (AI systems)
BMA	Bayesian Model Averaging
BN	Bayesian Network
CBN	Causal Bayesian Network
DAGGRE	Decomposition-based elicitation and AGGREgation
EA	Effective Altruism
FTX	FTX cryptocurrency exchange
IARPA	Intelligence Advanced Research Projects Activity
LLM	Large Language Model
PM-BN	Prediction Market–Bayesian Network
SBF	Sam Bankman-Fried
VCG	Vickrey-Clarke-Groves (mechanism)
VoI	Value of Information

D.2 Glossary of Key Terms

Term	Definition	Source
Cascade	Sequential decision-making where later actors rationally discount private information in favor of observed public signals	Bikhchandani et al. (1992, pp. 992–996)
Contamination	When model outputs influence forecasters whose subsequent forecasts are treated as independent evidence. Three mechanisms: (1) one-shot probabilistic anchoring, (2) feedback loop contamination producing variance collapse, (3) framing contamination through shared causal decompositions (Section II.5, Appendix A.5)	This paper
Epistemic hygiene	Practices that protect inferential validity against violations of independence assumptions	Pearl (2009)
Forecast quality	Combination of calibration (accuracy of confidence levels) and resolution (ability to discriminate outcomes)	Tetlock & Gardner (2015)
Hierarchical Bayesian	Multi-level probability model that explicitly represents observation processes and uncertainty at multiple scales	SEP

Term	Definition	Source
Noisy observation	A signal that imperfectly reflects underlying reality, requiring explicit modeling of the noise process	SEP
Proper scoring rule	A scoring function that incentivizes honest probability reporting (e.g., Brier score, logarithmic score)	Gneiting & Raftery (2007)
Spurious confirmation	Apparent independent validation that actually derives from the same original source	Pearl (2009)
Tacit ensemble knowledge	Experts' implicit distributions over multiple models, built through practice, that institutional formats systematically compress	Companion paper (PhilOfE-con)
Contamination signature	Multi-dimensional statistical distortion from contamination: mean bias, variance under/overestimation, dimensionality reduction, asymmetric/temporal effects	This paper (A.5)
Effective sample size	Adjusted sample size accounting for inter-forecaster correlation from shared AMTAIR exposure: $N_{\text{eff}} = (N - K) + K/(1 + (K - 1)\rho)$	This paper (A.5)
Uncertainty laundering	Process by which model uncertainty is obscured through circular feedback, returning as false confidence	This paper
Uncertainty preservation	Maintaining uncertainty in structured form until specific decisions force aggregation	Companion paper (PhilOfE-con)

D.3 List of Figures

1. **Figure 1** (Figure 1): DAGGRE vs AMTAIR Information Flow — Closed-loop vs open-loop architecture; dashed arrow marks unobservable contamination channel
2. **Figure 2** (Figure 2): Integration Architecture — Three-tier dependency: organizational infrastructure → parameter estimation → statistical framework
3. **Figure 3** (Figure 3): Three-Layer Hierarchical Bayesian Framework — Prior structure, forecast quality filtering (α , γ), and meta-integration

Appendix E: Cross-Paper Dependencies

This paper (V. J. Meyer, 2025c) is one of three companion papers exploring different dimensions of AMTAIR. Each can be read independently; this appendix traces the substantive connections.

Relation to the evolutionary epistemology paper (V. J. Meyer, 2025b). That paper provides the macro-level diagnosis this paper presupposes: AI governance is trapped in a pre-paradigmatic state where memetic fitness systematically outcompetes epistemic fitness. The cultural evolution framework (Sections III–IV of that paper) establishes *why* AMTAIR is needed --- selection pressures favor persuasiveness over precision. This paper takes that diagnosis as given and asks a different question: once AMTAIR exists, how should it integrate external data (prediction markets, forecasting platforms) without introducing

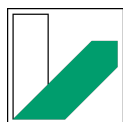
the very epistemic distortions it aims to correct? The contamination taxonomy developed here (Section III) feeds back into EvoRat's argument: the anti-gaming mechanisms proposed there (Section VI.2) gain specificity when contamination vectors are formally classified. EvoRat's Appendix H.6 traces the probabilistic dependencies in the other direction.

Relation to the philosophy of economics paper (V. J. Meyer, 2025d). That paper defends the epistemological foundations this paper's statistical framework operationalizes. PhilOfEcon argues that maintaining multiple competing models (model pluralism) is epistemically superior to premature model selection --- a claim this paper depends on when designing quality-weighted updating (Section IV.1) that preserves model diversity rather than collapsing to a single consensus. The level-separation framework in PhilOfEcon (epistemic, procedural, institutional) maps onto this paper's organizational decomposition: the analyst, team lead, and institutional layers (Section III) instantiate that abstract hierarchy. PhilOfEcon's Appendix B provides the Bayesian Model Averaging mathematics that underwrites this paper's ensemble approach.

What this paper contributes to the others. The contamination analysis is this paper's unique contribution to the thesis. EvoRat identifies cascade dynamics as the core pathology; this paper shows that AMTAIR's own infrastructure can *reproduce* those dynamics if prediction market data contaminates the models it aims to keep independent. The hierarchical Bayesian framework (Appendix A), embargo methodology (Section IV.2), and cascade detection metrics (Appendix A.4) provide the technical machinery that EvoRat's prediction tracking mechanism (Section IV.1) requires to function without self-undermining.

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Affidavit

Declaration of Academic Honesty

Hereby, I attest that I have composed and written the presented paper independently on my own, without the use of other than the stated aids and without any other resources than the ones indicated. All thoughts taken directly or indirectly from external sources are properly denoted as such.

This paper has neither been previously submitted in the same or a similar form to another authority nor has it been published yet.

BAYREUTH, December 31, 2025

VALENTIN MEYER